CHAPTER 1

INTRODUCTION

Depression has become a serious global mental-health concern, affecting more than 264 million people worldwide, and one of the groups most vulnerable to this condition includes students who struggle to secure employment after graduation. The transition from academic life to the competitive job market can be extremely stressful due to sudden lifestyle changes, high expectations, financial pressure, comparison with peers, and fear of an uncertain future, which may trigger or worsen depressive symptoms. These emotional changes may originate from alterations in the brain's neurotransmitter levels, stressful life events, or even genetic predispositions.

Many affected students suffer silently due to lack of awareness, stigma around mental illness, or limited access to mental-health support. As a result, untreated depression may lead to self-isolation, loss of confidence, poor concentration during job applications or interviews, dependency on medication, and in severe cases, suicidal thoughts. With the rapid growth of technology and the widespread adoption of social media—used by approximately 3.8 billion people worldwide —graduates often turn to online platforms to express their emotional struggles, frustrations, and personal experiences.

The text and posts shared on these platforms contain valuable insights that can be analyzed to understand the mental state of users. Therefore, social-media data can serve as an important resource for identifying early signs of depression among unemployed graduates, enabling timely support and intervention to improve their mental well-being and help them overcome career barriers. applications, people start to share their mental health battles with the world via social media instead of keeping it private. The data which is available in the user account can be analyzed to figure out their level of mental health and an opportunity to help them to recover.

1.1 Objectives

- The objective of this project is to develop an intelligent system capable of detecting early signs of depression in individuals using deep learning techniques.
- The system aims to analyze various data inputs such as facial expressions, voice tone, textual responses (like social media posts or chats), or questionnaire responses to identify potential depressive symptoms. By leveraging deep learning models like Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), or transformer-based models, the goal is to achieve high accuracy in identifying emotional patterns indicative of depression.
- This can support early diagnosis and timely intervention, ultimately helping mental health professionals in providing better care.
- The objective of this study is to investigate the feasibility of detecting depression from user tweets using machine learning techniques. By analyzing a dataset comprising Twitter posts, we aim to develop a robust and accurate predictive model capable of identifying users who may be at risk of depression. This research can contribute to the development of innovative digital health tools that facilitate early intervention and provide support to individuals in need of mental health assistance.
- Financial and time constraints often limit the creation of dedicated datasets for depression detection. The study proposes to bridge the gap by repurposing an existing dataset originally designed for sentiment analysis.

CHAPTER 2

LITERATURE REVIEW

1. Depression detection from social network data using machine learning techniques Md. Rafiqul Islam, Muhammad Ashad Kabir, Ashir Ahmed, Abu Raihan M. Kamal, Hua Wang, and Anwaar Ulhaq (2018).

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6111060/

ABSTRACT: Purpose Social networks have been developed as a great point for its users to communicate with their interested friends and share their opinions, photos, and videos reflecting their moods, feelings and sentiments. This creates an opportunity to analyze social network data for user's feelings and sentiments to investigate their moods and attitudes when they are communicating via these online tools. Methods Although diagnosis of depression using social networks data has picked an established position globally, there are several dimensions that are yet to be detected. In this study, we aim to perform depression analysis on Facebook data collected from an online public source. To investigate the effect of depression detection, we propose machine learning technique as an efficient and scalable method. Results We report an implementation of the proposed method. We have evaluated the efficiency of our proposed method using a set of various psycholinguistic features. We show that our proposed method can significantly improve the accuracy and classification error rate. In addition, the result shows that in different experiments Decision Tree (DT) gives the highest accuracy than other ML approaches to find the depression. Conclusions Machine learning techniques identify high quality solutions of mental health problems among Facebook users.

2. Detecting Depression Using K-Nearest Neighbors (KNN) Classification Technique — Md. Rafiqul Islam, Abu Raihan M. Kamal, Naznin Sultana, Robiul Islam, Mohammad Ali Moni, and Anwaar Ulhaq (2018).

https://ieeexplore.ieee.org/document/8465641

ABSTRACT: Social networks have developed as a promising point for everybody to communicate with their interested friend and share their opinions, photos, and videos. Also, it has been an upcoming research field and has picked an established position globally. In this paper, we considered depression problems among various Facebook users. Already, a number of researchers have studied and applied many techniques to detect depression, but still need to detect accurately from social network data. So, we investigate the possibility to utilize Facebook data and apply KNN (k-nearest neighbors) classification technique for detecting depressive emotions. We do believe that our investigation and approach might be helpful to raise consciousness in online social network users.

3. Machine Learning-based Approach for Depression Detection in Twitter Using Content and Activity Features -- Hatoon AlSagri and Mourad Ykhlef (2020)

https://arxiv.org/abs/2003.04763

ABSTRACT: Social media channels, such as Facebook, Twitter, and Instagram, have altered our world forever. People are now increasingly connected than ever and reveal a sort of digital persona. Although social media certainly has several remarkable features, the demerits are undeniable as well. Recent studies have indicated a correlation between high usage of social media sites and increased depression. The present study aims to exploit machine learning techniques for detecting a probable depressed Twitter user based on both, his/her network behavior and tweets. For this purpose, we trained and tested classifiers to distinguish whether a user is depressed or not using features extracted from his/ her activities in the network and tweets. The results showed that the more features are used, the higher are the accuracy and F-measure scores in detecting depressed users. This method is a data-driven, predictive approach for early detection of depression or other mental illnesses. This study's main contribution is the exploration part of the features and its impact on detecting the depression level.

4. Depression Detection and Prevention System by Analysing Tweets — Mrunal Gaikar, Jayesh Chavan, Kunal Indore, and Rajashree Shedge (2019)

https://link.springer.com/article/10.1007/s40860-022-00171-8

ABSTRACT: Social media platforms like Twitter which is a microblogging tool enable its users to express their feelings, emotions and opinions through short text messages. Detecting the emotions in a text can help one identify anxiety and depression of an individual. Depression is a mental health problem which can happen to anyone, at any age. There is a lack of systematic and efficient methods to identify the psychological state of an individual. With more than 58 millions tweets generated daily, Twitter can be used in order to detect the sign of depression in a faster way. Recent studies have demonstrated that Twitter can be used to prevent one from taking an extreme step. Our Proposed depression detection and prevention system can detect any depression related words or phrases from Tweets and also classify the type of depression, if detected. This system is proposed in order to diagnose depression and prevent it. Proposed system is using Support vector machine and Naïve Bayes classifier. This hybrid approach works well not only with shorter snippets but also with longer snippets.

5. Predicting Depression Levels Using Social Media Posts — Maryam Mohammed Aldarwish and Hafiz Farooq Ahmad (2017)

https://ieeexplore.ieee.org/document/7940253

ABSTRACT: The use of Social Network Sites (SNS) is increasing nowadays especially by the younger generations. The availability of SNS allows users to express their interests, feelings and share daily routine. Many researchers prove that using user-generated content (UGC) in a correct way may help determine people's mental health levels. Mining the UGC could help to predict the mental health levels and depression. Depression is a serious medical illness, which interferes most with the ability to work, study, eat, sleep and having fun. However, from the user profile in SNS, we can collect all the information that relates to person's mood, and negativism. In this research, our aim is to investigate how SNS user's posts can help classify users according to mental health levels. We propose a system that uses SNS as a source of data and screening tool to classify the user using artificial intelligence according to the UGC on SNS. We created a model that classify the UGC using two different classifiers: Support Vector Machine (SVM), and Naïve Bayes.

6. F. Sadeque, D. Xu, and S. Bethard, “Measuring Latency of Depression Detection in Social Media.” WSDM'18, February 5–9, 2018, Marine Del Rey, CA, USA, 2018.

ABSTRACT: Detecting depression is a key public health challenge, as almost 12% of all disabilities can be attributed to depression. Computational models for depression detection must prove not only

that can they detect depression, but that they can do it early enough for an intervention to be plausible. However, current evaluations of depression detection are poor at measuring model latency. We identify several issues with the currently popular ERDE metric, and propose a latency-weighted F1 metric that addresses these concerns. We then apply this evaluation to several models from the recent eRisk 2017 shared task on depression detection, and show how our proposed measure can better capture system differences.

7. D. E. Losada and F. Crestani, “A Test Collection for Research on Depression and Language Use ”.

ABSTRACT: International Conference of the Cross-Language Evaluation Forum for European Languages. Springer International Publishing., 2016. Several studies in the literature have shown that the words people use are indicative of their psychological states. In particular, depression was found to be associated with distinctive linguistic patterns. However, there is a lack of publicly available data for doing research on the interaction between language and depression. In this paper, we describe our first steps to fill this gap. We outline the methodology we have adopted to build and make publicly available a test collection on depression and language use. The resulting corpus includes a series of textual interactions written by different subjects. The new collection not only encourages research on differences in language between depressed and non-depressed individuals, but also on the evolution of the language use of depressed individuals. Further, we propose a novel early detection task and define a novel effectiveness measure to systematically compare early detection algorithms. This new measure takes into account both the accuracy of the decisions taken by the algorithm and the delay in detecting positive cases. We also present baseline results with novel detection methods that process users’ interactions in different ways.

8. T. Nguyen, D. Phung, S. V. B. Dao, and M. Berk, “Affective and content analysis of online depression communities.” IEEE Trans. Affect. Comput., vol. 5, no. 3, pp. 217–226, 2014.

ABSTRACT: A large number of people use online communities to discuss mental health issues, thus offering opportunities for new understanding of these communities. This paper aims to study the characteristics of online depression communities (CLINICAL) in comparison with those joining other online communities (CONTROL). We use machine learning and statistical methods to discriminate online messages between depression and control communities using

mood, psycholinguistic processes and content topics extracted from the posts generated by members of these communities. All aspects including mood, the written content and writing style are found to be significantly different between two types of communities. Sentiment analysis shows the clinical group have lower valence than people in the control group. For language styles and topics, statistical tests reject the hypothesis of equality on psycholinguistic processes and topics between two groups. We show good predictive validity in depression classification using topics and psycholinguistic clues as features. Clear discrimination between writing styles and contents, with good predictive power is an important step in understanding social media and its use in mental health.

9. Fariq Sadeque and Dongfang Xu and Steven Bethard, "UArizona at the CLEF eRisk 2017 Pilot Task: Linear and Recurrent Models for Early Depression Detection ",

ABSTRACT: CEUR Workshop Proc. Author manuscript; PMC 2017. The 2017 CLEF eRisk pilot task focuses on automatically detecting depression as early as possible from a users' posts to Reddit. In this paper we present the techniques employed for the University of Arizona team's participation in this early risk detection shared task. We leveraged external information beyond the small training set, including a preexisting depression lexicon and concepts from the Unified Medical Language System as features. For prediction, we used both sequential (recurrent neural network) and non sequential (support vector machine) models. Our models perform decently on the test data, and the recurrent neural models perform better than the non-sequential support vector machines while using the same feature sets.

10. Schuster, Mike Paliwal, Kuldip. "Bidirectional recurrent neural networks.", Signal Processing, IEEE Transactions on. 45. 2673–2681. 10.1109/78.650093, 1997.

ABSTRACT: In the first part of this paper, a regular recurrent neural network (RNN) is

extended to a bidirectional recurrent neural network (BRNN). The BRNN can be trained without the limitation of using input information just up to a preset future frame. This is accomplished by training it simultaneously in positive and negative time direction. Structure and training procedure of the proposed network are explained. In regression and classification experiments
on artificial data, the proposed structure gives better results than other approaches. For real data, classification experiments for phonemes from the TIMIT database show the same tendency. In the second part of this paper, it is shown how the proposed bidirectional structure can be easily modified to allow efficient estimation of the conditional posterior probability of

complete symbol sequences without making any explicit assumption about the shape of the distribution. For this part, experiments on real data are reported.

11. Trotzek, M., Koitka, S., Friedrich, and C.M., "Linguistic Metadata Augmented Classifiers at the CLEF 2017 Task for Early Detection of Depression". Working Notes Conference and Labs of the Evaluation Forum CLEF 2017, Dublin, Ireland, 2017.

ABSTRACT: Methods for automatic early detection of depressed individuals based on written texts can help in research of this disorder and especially offer better assistance to those affected. FHDO Biomedical Computer Science Group (BCSG) has submitted results obtained from five models for the CLEF 2017 eRisk task for early detection of depression that are described in this paper. All models utilize linguistic meta information extracted from the texts of each evaluated user and combine them with classifiers based on Bag of Words (BoW) models, Paragraph Vector, Latent Semantic Analysis (LSA), and Recurrent Neural Networks (RNN) using Long Short Term Memory (LSTM). BCSG has achieved top performance according to ERDE5 and F1 score for this task. 3.12 Diederik Kingma and Jimmy Ba. "Adam:A method for stochastic opti- mization.Intemational Conference on Learning Representation " We introduce Adam, an algorithm for first-order gradient-based optimization of stochastic objective functions, based on adaptive estimates of lower-order moments. The method is straightforward to implement, is computationally efficient, has little memory requirements, is invariant to diagonal rescaling of the gradients, and is well suited for problems that are large in terms of data and/or parameters. The method is also appropriate for non-stationary objectives and problems with very noisy and/or sparse gradients. The hyperparameters have intuitive interpretations and typically require little tuning. Some connections to related algorithms, on which Adam was inspired, are discussed. We also analyze the theoretical convergence properties of the algorithm and provide a regret bound on the convergence rate that is comparable to the best known results under the online convex optimization framework. Empirical results demonstrate that Adam works well in practice and compares favorably to other stochastic optimization methods. Finally, we discuss AdaMax, a variant of Adam based on the infinity norm.

13. Tausczik, Y.R., Pennebaker, J.W. "The Psychological Meaning of Words: LIWC and Computerized Text Analysis Methods. Journal of Language and Social Psychology ", Vol. 29 (1), pp. 24–54. 2010.

ABSTRACT: We are in the midst of a technological revolution whereby, for the first time, researchers can link daily word use to a broad array of real-world behaviors. This article reviews several computerized text analysis methods and describes how Linguistic Inquiry and Word Count (LIWC) was created and validated. LIWC is a transparent text analysis program that counts words in psychologically meaningful categories. Empirical results using LIWC demonstrate its ability to detect meaning in a wide variety of experimental settings, including to show attentional focus, emotionality, social relationships, thinking styles, and individual differences.

14. A. Vázquez-Romero and A. Gallardo-Antolín, "Automatic Detection of Depression in Speech Using Ensemble Convolutional Neural Networks," Entropy, vol. 22, no. 6, p. 688, Jun. 2020.

ABSTRACT: This paper proposes a speech-based method for automatic depression classification. The system is based on ensemble learning for Convolutional Neural Networks (CNNs) and is evaluated using the data and the experimental protocol provided in the Depression Classification Sub-Challenge (DCC) at the 2016 Audio–Visual Emotion Challenge (AVEC- 2016). In the pre-processing phase, speech files are represented as a sequence of log-spectrograms and randomly sampled to balance positive and negative samples. For the classification task itself, first, a more suitable architecture for this task, based on One-Dimensional Convolutional Neural Networks, is built. Secondly, several of these CNN-based models are trained with different initializations and then the corresponding individual predictions are fused by using an Ensemble Averaging algorithm and combined per speaker to get an appropriate final decision. The proposed ensemble system achieves satisfactory results on the DCC at the AVEC-2016 in comparison with a reference system based on Support Vector Machines and hand-crafted features, with a CNN+LSTM-based system called DepAudionet, and with the case of a single CNN-based classifier.

15. S. Krishna and Anju J, "Speech Based Depression Detection using Convolution Neural Networks," Regular, vol. 9, no. 9, pp. 405–408, 2020.

ABSTRACT: Automatic diagnosis of depression based on speech can complement mental health treatment methods in the future. Previous studies have reported that acoustic properties can be used to identify depression. However, few studies have attempted a large-scale differential diagnosis of patients with depressive disorders using acoustic characteristics of non-English speakers.

16. K. Chlasta, K. Wołk and I. Krejtz, "Automated speechbased screening of depression using deep convolutional neural networks", Procedia Computer Science, vol. 164, pp. 618-628, 2019.

ABSTRACT: Early detection and treatment of depression is essential in promoting remission, preventing relapse, and reducing the emotional burden of the disease. Current diagnoses are primarily subjective, inconsistent across professionals, and expensive for individuals who may be in urgent need of help. This paper proposes a novel approach to automated depression detection in speech using convolutional neural network (CNN) and multipart interactive training. The model was tested using 2568 voice samples obtained from 77 non-depressed and 30 depressed individuals. In experiment conducted, data were applied to residual CNNs in the form of spectrograms—images auto-generated from audio samples. The experimental results obtained using different ResNet architectures gave a promising baseline accuracy reaching 77%.

17. J. R. Williamson, T. F. Quatieri, B. S. Helfer, R. Horwitz, B. Yu, and D. D. Mehta, "Vocal biomarkers of depression based on motor incoordination," Proceedings of the 3rd ACM international workshop on Audio/visual emotion challenge, Oct. 2013.

ABSTRACT: In Major Depressive Disorder (MDD), neurophysiologic changes can alter motor control [1, 2] and therefore alter speech production by influencing the characteristics of the vocal source, tract, and prosodics. Clinically, many of these characteristics are associated with psychomotor retardation, where a patient shows sluggishness and motor disorder in vocal articulation, affecting coordination across multiple aspects of production [3, 4]. In this paper, we exploit such effects by selecting features that reflect changes in coordination of vocal tract motion associated with MDD.

Specifically, we investigate changes in correlation that occur at different time scales across formant frequencies and also across channels of the delta-mel- cepstrum. Both feature domains provide measures of coordination in vocal tract articulation while reducing effects of a slowly-varying linear channel, which can be introduced by time- varying microphone placements. With these two complementary feature sets, using the AVEC 2013 depression dataset, we design a novel Gaussian mixture model (GMM)-based multivariate regression scheme, referred to as Gaussian Staircase Regression, that provides a root-mean-squared-error (RMSE) of 7.42 and a mean absolute-error (MAE) of 5.75 on the standard Beck depression rating scale. We are currently exploring coordination measures of other aspects of speech production, derived from both audio and video signals.

18. S. Dham, A. Sharma, and A. Dhall, "Depression scale recognition from audio, visual and text analysis," arXiv preprint, arXiv:1709.05865, 2017.

ABSTRACT: Depression is a major mental health disorder that is rapidly affecting lives worldwide. Depression not only impacts emotional but also physical and psychological state of the person. Its symptoms include lack of interest in daily activities, feeling low, anxiety, frustration, loss of weight and even feeling of self hatred. This report describes work done by us for Audio Visual Emotion Challenge (AVEC) 2017 during our second year BTech summer internship. With the increase in demand to detect depression automatically with the help of machine learning algorithms, we present our multimodal feature extraction and decision level fusion approach for the same. Features are extracted by processing on the provided Distress

Analysis Interview Corpus-Wizard of Oz (DAIC-WOZ) database. Gaussian Mixture Model (GMM) clustering and Fisher vector approach were applied on the visual data; statistical descriptors on gaze, pose; low level audio features and head pose and text features were also extracted. Classification is done on fused as well as independent features using Support Vector Machine (SVM) and neural networks. The results obtained were able to cross the provided baseline on validation data set by 17% on audio features and 24.5% on video features.

19. J. F. Cohn et al., "Detecting depression from facial actions and vocal prosody," 2009 3rd International Conference on Affective Computing and Intelligent Interaction and Workshops, Amsterdam, Netherlands, 2009, pp. 1 7, doi: 10.1109/ACII.2009.5349358.

ABSTRACT: Current methods of assessing psychopathology depend almost entirely on verbal report (clinical interview or questionnaire) of patients, their family, or caregivers. They lack systematic and efficient ways of incorporating behavioral observations that are strong indicators of psychological disorder, much of which may occur outside the awareness of either individual. We compared clinical diagnosis of major depression with automatically measured facial actions and vocal prosody in patients undergoing treatment for depression. Manual FACS coding, active appearance modeling (AAM) and pitch extraction were used to measure facial and vocal expression. Classifiers using leave-one-out validation were SVM for FACS and for AAM and logistic regression for voice. Both face and voice demonstrated moderate concurrent validity with depression. Accuracy in detecting depression was 88% for manual FACS and 79% for AAM. Accuracy for vocal prosody was 79%. These findings suggest the feasibility of automatic detection of depression, raise new issues in automated.

2.20 Tong, Y., Liao, W., and Ji, Q. : „ Facial action unit recognition by exploiting their dynamic and semantic relationships,‟ IEEE Transactions on Pattern Analysis and Machine Intelligence, 2007, 29, (10), pp. 1683–1699.

ABSTRACT: A system that could automatically analyze the facial actions in real time has applications in a wide range of different fields. However, developing such a system is always challenging due to the richness, ambiguity, and the dynamic nature of facial actions. Although a number of research groups attempt to recognize facial action units (AUs) by either improving facial feature extraction techniques, or the AU classification techniques, these methods often recognize AUs or certain AU combinations individually and statically, ignoring the semantic relationships among AUs and the dynamics of AUs. Hence, these approaches cannot always recognize AUs reliably, robustly, and consistently.In this paper, we propose a novel approach that systematically accounts for the relationships among AUs and their temporal evolutions for AU recognition. Specifically, we use a dynamic Bayesian network (DBN) to model the relationships among different AUs. The DBN provides a coherent and unified hierarchical probabilistic framework to represent probabilistic relationships among various AUs and to account for the temporal changes in facial action development.

21. Lucey, P., Cohn, J.F., Lucey, S., Sridharan, S., and Prkachin, K. : „ Automatically detecting action units from faces of pain: Comparing shape and appearance features "Proc. 2nd IEEE Workshop on CVPR for Human Communicative Behavior Analysis (CVPR4HB), June 2009.

ABSTRACT: Recent psychological research suggests that facial movements are a reliable measure of pain. Automatic detection of facial movements associated with pain would contribute to patient care but is technically challenging. Facial movements may be subtle and accompanied by abrupt changes in head orientation. Active appearance models (AAM) have proven robust to naturally occurring facial behavior, yet AAM based efforts to automatically detect action units (AUs) are few. Using image data from patients with rotator-cuff injuries, we describe an AAM-based automatic system that decouples shape and appearance to detect AUs on a frame-by-frame basis. Most current approaches to AU detection use only appearance features. We explored the relative efficacy of shape and appearance for AU detection. Consistent with the experience of human observers, we found specific relationships between action units and types of facial features. Several AU (e.g. AU4, 12, and 43) yield optimal results.

22. Valstar, M.F., Pantic, M., Ambadar, Z., and Cohn, J.F. : „ Spontaneous vs. posed facial behavior: Automatic analysis of brow actions,". Proc. ACM International Conference on Multimodal Interfaces, Banff, Canada, November 2006.

ABSTRACT: Past research on automatic facial expression analysis has focused mostly on the recognition of prototypic expressions of discrete emotions rather than on the analysis of dynamic changes over time, although the importance of temporal dynamics of facial expressions for interpretation of the observed facial behavior has been acknowledged for over 20 years. For instance, it has been shown that the temporal dynamics of spontaneous and volitional smiles are fundamentally different from each other. In this work, we argue that the same holds for the temporal dynamics of brow actions and show that velocity, duration, and order of occurrence of brow actions are highly relevant parameters for distinguishing posed from spontaneous brow actions. The proposed system for discrimination between volitional and spontaneous brow actions is based on automatic detection of Action Units (AUs) and their temporal segments (onset, apex, offset) produced by movements of the eyebrows. For each temporal segment of an activated AU, we compute a number of mid-level feature parameters including the maximal intensity, duration, and order of occurrence.

23. De la Torre, F. Campoy, J. Ambadar, Z. and Cohn, J.F. : „ Temporal segmentation of facial behavior,‟ IEEE International Conference on Computer Vision, 2007, pp. xxx–xxx

ABSTRACT: Temporal segmentation of facial gestures in spontaneous facial behavior recorded in real- world settings is an important, unsolved, and relatively unexplored problem in facial image analysis. Several issues contribute to the challenge of this task. These include non-frontal pose, moderate to large out-of-plane head motion, large variability in the temporal scale of facial gestures, and the exponential nature of possible facial action combinations. To address these challenges, we propose a two step approach to temporally segment facial behavior. The first step uses spectral graph techniques to cluster shape and appearance features invariant to some geometric transformations. The second step groups the clusters into temporally coherent facial gestures. We evaluated this method in facial behavior recorded during face-to- face interactions. The video data were originally collected to answer substantive questions in psychology without concern for algorithm development. The method achieved moderate convergent validity with manual FACS (Facial Action Coding System) annotation. Further, when used to preprocess video for manual FACS annotation, detect unusual facial behavior.

24. Nikhil Marriwala, Deepti Chaudhary Measurement: Sensors 25, 100587, 2023

ABSTRACT: Millions of people are suffering from mental illness due to unavailability of early treatment and services for depression detection. It is the major reason for anxiety disorder, bipolar disorder, sleeping disorder, depression and sometimes it may lead to self-harm and suicide. Thus, it is a very challenging task to recognize people who are suffering from mental health disorders and provide them treatments as early as possible. Conventionally, depression detection was done through patients' interviews and PHQ scores, accuracy of conventional methods is very less. In this work, Bi-LSTM model is also used in the proposed work. In results, training accuracy, training loss, F1-score, recall and support are found for evaluation of models. In results, graphs for training loss, validation loss, training accuracy and validation accuracy are plotted. At last, by using confusion matrix depression can be detected for textual CNN Model, audio CNN model, LSTM model and Bi-LSTM against true label and predicted label.

25. Wadzani Aduwamai Gadzama, Danlami Gabi, Musa Sule Argungu, Hassan Umar Suru Personalized Medicine in Psychiatry 45, 100125, 2024

ABSTRACT: Depression is regarded as one of the world's primary concerns. Recent researchers use artificial intelligence techniques like machine learning and deep learning to identify depressive symptoms automatically. This literature review focuses on using machine learning and deep learning models in depression detection on social media. Advances in deep learning have improved methods for identifying depression, which is one of the illnesses that affect the health of individuals. Some researchers employ a variety of deep-learning approaches to.

26. Indira Sen, Daniele Quercia, Marios Constantinides, Matteo Montecchi, Licia Capra, Sanja Scepanovic, Renzo Bianchi Proceedings of the ACM on Human-Computer Interaction 6 (CSCW2), 1-21, 2022

ABSTRACT: Studies on depression in the workplace have mostly investigated its impact on individual employees. Little is known about its association with the company as a whole, or the state where the company is based. This is due to the lack of scalable methodologies

operationalizing depression in the specific context of the workplace, and of data documenting potential distress.
In this work, but also at macro-level (states hosting these companies were associated with high depression rates, talent shortage, and economic deprivation). This new way of applying AutoODI onto company reviews offers both theoretical implications for the literature in computational social science, occupational health and economic geography, and practical implications for companies and policy makers.

27. Jiayu Ye, et al., (2021), "Multi-modal depression detection based on emotional audio and evaluation text", Journal of Affective Disorders, Volume 295, DOI: 10.1016/j.jad.2021.08.090

ABSTRACT: Unemployment ranks among the most important socioeconomic drivers of health disparities in terms of its effect on people's chances of living longer and healthier lives. The relationship between unemployment and ill health is the result of the interaction of several selection and causative factors. Investigating this intricate link in depth, researchers have shown that it affects people's lives in a variety of ways, from their financial security to their susceptibility to mental and physical disorders. Depression stands out as a particularly prevalent and closely related outcome when examining how unemployment affects an individual's mental health.

28. Nazmun Nessa Moon, et al., (2021), "Machine learning approach to predict the depression in job sectors in Bangladesh", Current Research in Behavioral Sciences, Volume 2, DOI: 10.1016/j.crbeha.2021.100058

ABSTRACT: The leadership and culture of the company have a big impact on the mental health of its employees. A toxic work environment, which is defined by a lack of transparency, poor communication, and a severe response to mistakes, may make workplace depression worse. Leadership styles that prioritize efficiency over employee satisfaction, minimize the importance of emotional health, or promote healthy competition over teamwork can all contribute to workplace difficulties. On the other hand, organizations that respect and encourage an open and supportive culture, employee well-being initiatives, and open communication are less likely to experience workplace depression.

29. Lei, L., et al., (2023), "Assessing the role of artificial intelligence in the mental healthcare of teachers and students". Soft Comput. DOI: 10.1007/s00500-023-08072-5

ABSTRACT: Unbalanced workloads and a lack of social support are two prevalent factors that exacerbate work-related depression. Long hours frequently cause workers to fall into a rut, leaving them with little time for their family and themselves. In particular, the rise of remote work has made it harder to distinguish between work and personal life, which has been connected to higher stress levels and an incessant desire to be "on." Lack of a strong support system at work might make people feel even more alone because they may be reluctant to voice their problems for fear of criticism or repercussions.

30. Rokeya Siddiqua, et al., (2023), "AIDA: Artificial intelligence based depression assessment applied to Bangladeshi students", Array, Volume 18, DOI: 10.1016/j.array.2023.

ABSTRACT: 100291 The prevalence of depression is sharply increasing in developing countries such as Bangladesh. In most cases, a timely diagnosis is beneficial when treating those who may be depressed. In order to develop methods for detecting and forecasting depression levels, focuses on college students in Bangladesh. They created a survey with 106 questions. The survey consists of two primary question types: (i) objective clinical questions and (ii) introspective questions. 684 responses from Bangladeshi students aged 19 to 35 who were requested to complete the survey were gathered. Only after completing the necessary permissions were participants granted access to the survey. For this sample, 520 responses in total were thoroughly reviewed. Eight well-known depression evaluation tools were merged into one hybrid scale through a vote process. This hybrid scale was applied to the collected data, which included questions from several widely used depression assessment tools as well as sensitive information. Ten AI models were also used to forecast the three levels of depression: normal, severe, and excessive. Nine feature selection methods and five hyper parameter optimizers were used to increase the predictability. The most accurate models were found to be RF models (98.08%), Gradient Boosting (94.23%), and Convolutional Neural Networks (92.31%). The best F Measure and the fewest false negatives were obtained with the RF hyperparameters optimized.

31. Joshi, et al., (2022), "Depression detection using emotional artificial intelligence and machine learning: A closer review" Materials Today: Proceedings. 58. DOI: 10.1016/j.matpr.2022.01.467

ABSTRACT: The use of electroencephalography (EEG) to diagnose mental illnesses like depression is currently quite popular due to developments in biomedical technology. Two significant challenges to this application are the intricacy and non-stationarity of EEG waves.
Furthermore, impacts caused by individual variances may hinder the generalization of detection systems. This project's main objective is to develop an algorithm that can recognize depressed symptoms using EEG data analysis. Following a multiband analysis of such signals, machine learning and deep learning techniques were applied to automatically detect depressed people.

CHAPTER 3

FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBILITY

3.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

3.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

3.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER 4

SYSTEM ANALYSIS

4.1 EXISTING SYSTEM:

In existing they study addresses the pressing issue of early risk detection for depression and anorexia through the analysis of text data from social media platforms such as Twitter and Reddit, as demonstrated in the context of the CLEF eRisk 2018 competition. With the increasing prominence of social media as a communication medium, it has become a valuable source for identifying potential mental health risks. Our system approach combines the power of TF-IDF information and convolutional neural networks (CNNs) to effectively identify text content indicative of depression and anorexia. The results of our evaluation exhibit promising performance, with an ERDE5 of 10.81%, ERDE50 of 9.22%, and F-score of 0.37 in depression detection, and an ERDE5 of 13.65%, ERDE50 of 11.14%, and F-score of 0.67 in anorexia detection. This research represents a significant step towards leveraging social media data for proactive mental health risk identification.

4.1.1 DISADVANTAGES OF EXISTING SYSTEM:

- The existing work primarily analyzes text data from social media platforms.
- The existing work reports performance metrics like ERDE5, ERDE50, and F-score for depression and anorexia detection.
- The existing work does not mention specific measures for privacy and data security.

4.2 Proposed System:

The proposed system in this study aims to create an efficient and accurate framework for the early detection of depression using machine learning. This system will leverage a diverse ensemble of machine learning algorithms, including Naive Bayes, Support Vector Machines (SVM), K-Nearest Neighbors (KNN), Random Forest, Bagging, Boosting, Neural Network, Logistic Regression, XGBoost, Perceptron, and a Voting Classifier. These algorithms will be employed to analyze and interpret various data sources, such as text-based social media content, physiological indicators, and self-reported questionnaires, to identify potential indicators of depression. The system will integrate these algorithms into a unified platform, allowing for the automated screening and assessment of depression risk in individuals. It will provide healthcare professionals and mental health practitioners with a valuable tool for early intervention and personalized treatment planning. Additionally, the proposed system will prioritize privacy and data security, ensuring the confidentiality of sensitive information. This research represents a significant step towards harnessing the power of machine learning to improve mental health outcomes by enabling timely and accurate depression detection.

4.2.1 Advantages of proposed system:

1. The present work not only analyzes text-based social media content but also incorporates physiological indicators and self-reported questionnaires, providing a more comprehensive approach to depression detection.
2. The present work doesn't provide specific performance metrics but aims to create an "efficient and accurate framework," suggesting a commitment to improving performance.
3. The present work emphasizes the importance of privacy and data security, ensuring the confidentiality of sensitive information, which is crucial when dealing with mental health data.

4.3 FUNCTIONAL REQUIREMENTS

1.Data Collection

2.Data Preprocessing

3.Training And Testing

4.Modiling

5.Predicting

4.4 NON FUNCTIONAL REQUIREMENTS

NON-FUNCTIONAL REQUIREMENT (NFR) specifies the quality attribute of a software system. They judge the software system based on Responsiveness, Usability, Security, Portability and other non-functional standards that are critical to the success of the software system. Example of nonfunctional requirement, “how fast does the website load?” Failing to meet non-functional requirements can result in systems that fail to satisfy user needs. Non-functional Requirements allows you to impose constraints or restrictions on the design of the system across the various agile backlogs. Example, the site should load in 3 seconds when the number of simultaneous users are > 10000. Description of non-functional requirements is just as critical as a functional requirement.

- Usability requirement
- Serviceability requirement
- Manageability requirement
- Recoverability requirement
- Security requirement
- Data Integrity requirement
- Capacity requirement
- Availability requirement
- Scalability requirement
- Interoperability requirement
- Reliability requirement
- Maintainability requirement
- Regulatory requirement
- Environmental requirement

CHAPTER 5

METHODOLOGY

5.1 SYSTEM ARCHITECTURE:

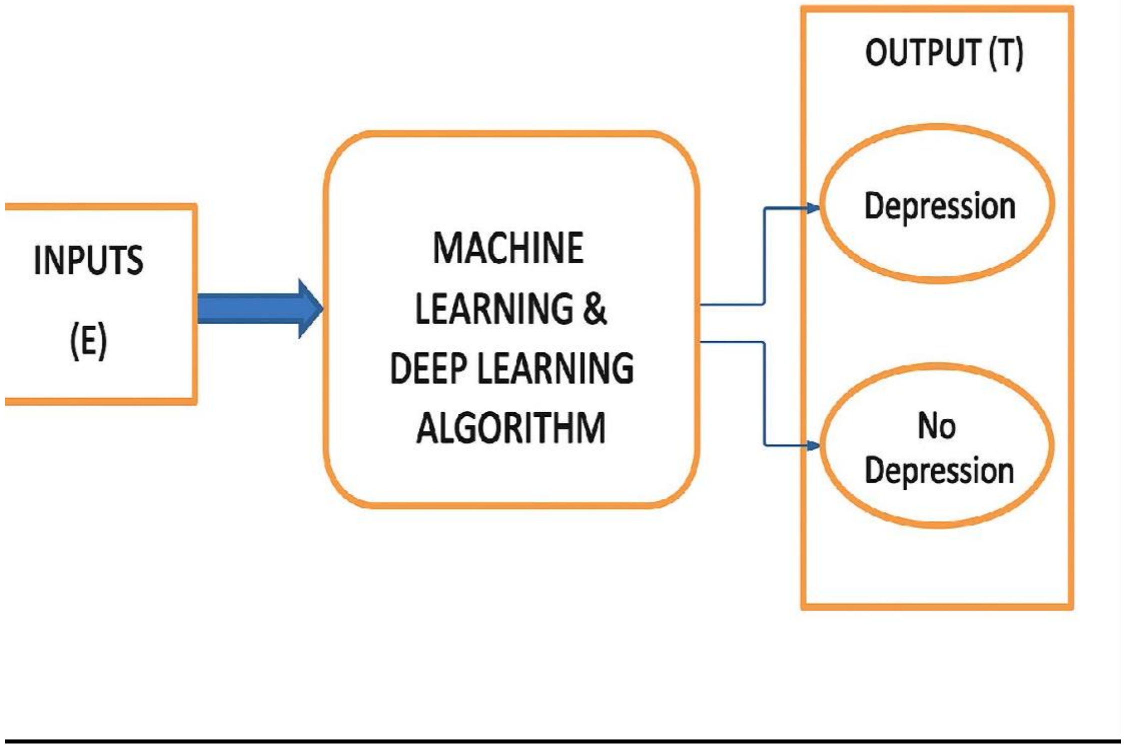

Fig: 5.1.1System Architecture

5.1.2 DATA FLOW DIAGRAM:

1. The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
3. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.

4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

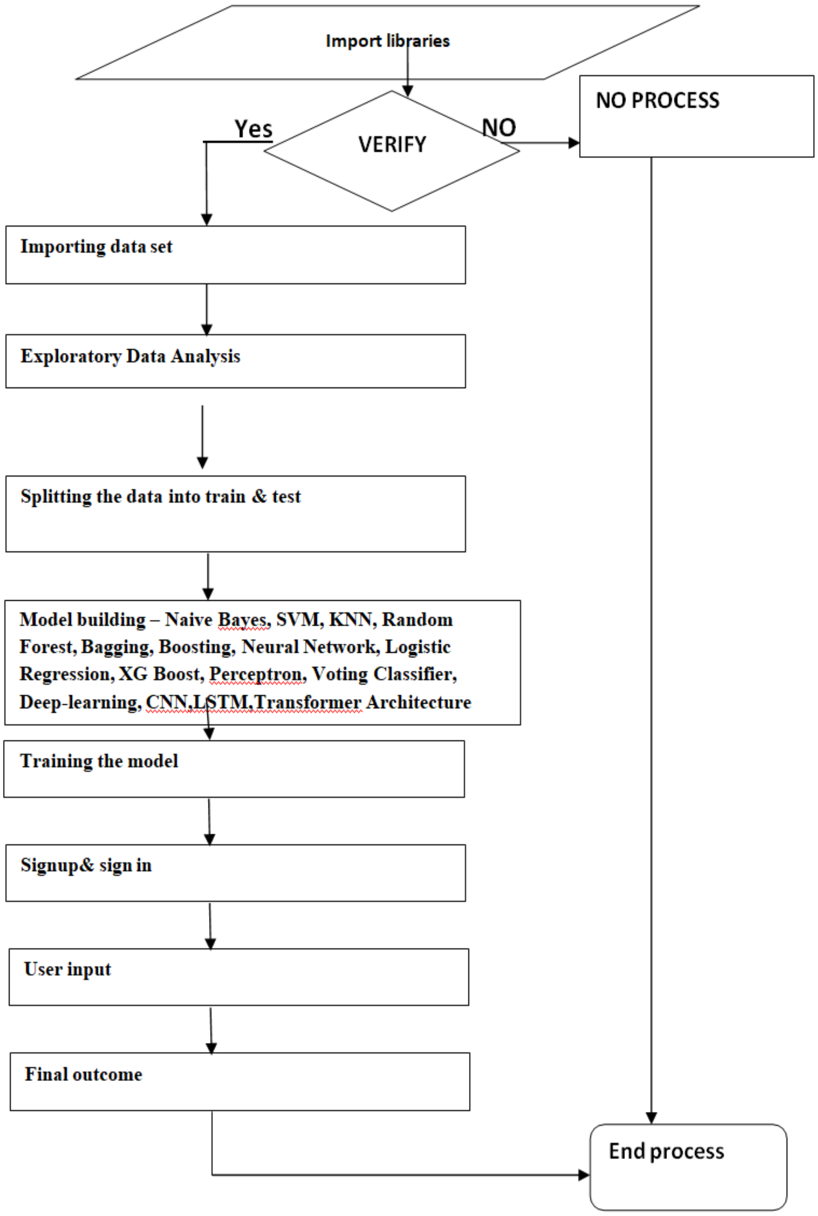

Fig.5.1.2 Dataflow diagram

5.2 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development process.
4. Provide a formal basis for understanding the modeling language.
5. Encourage the growth of OO tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns and components.Integrate best practices.

5.2.1 Use case diagram:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

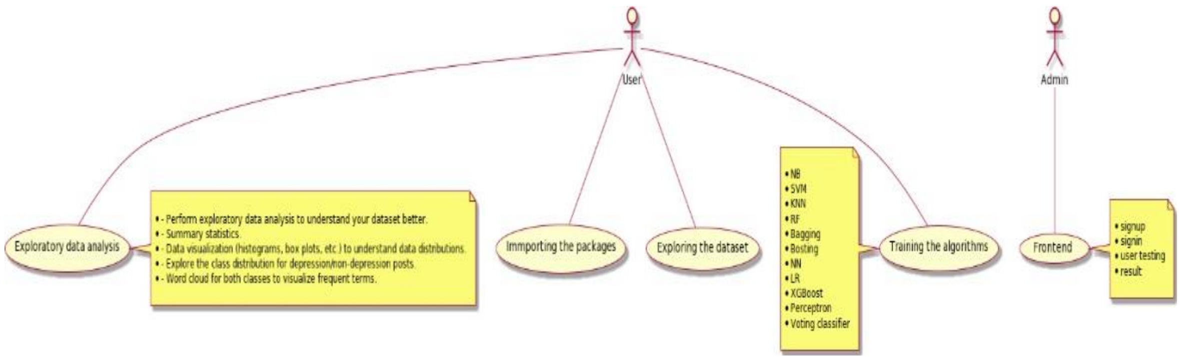

Fig.5.2.1 Use case diagram

5.2.2 Class diagram:

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an "is-a" or "has-a" relationship. Each class in the class diagram may be capable of providing certain functionalities. These functionalities provided by the class are termed "methods" of the class. Apart from this, each class may have certain "attributes" that uniquely identify the class.

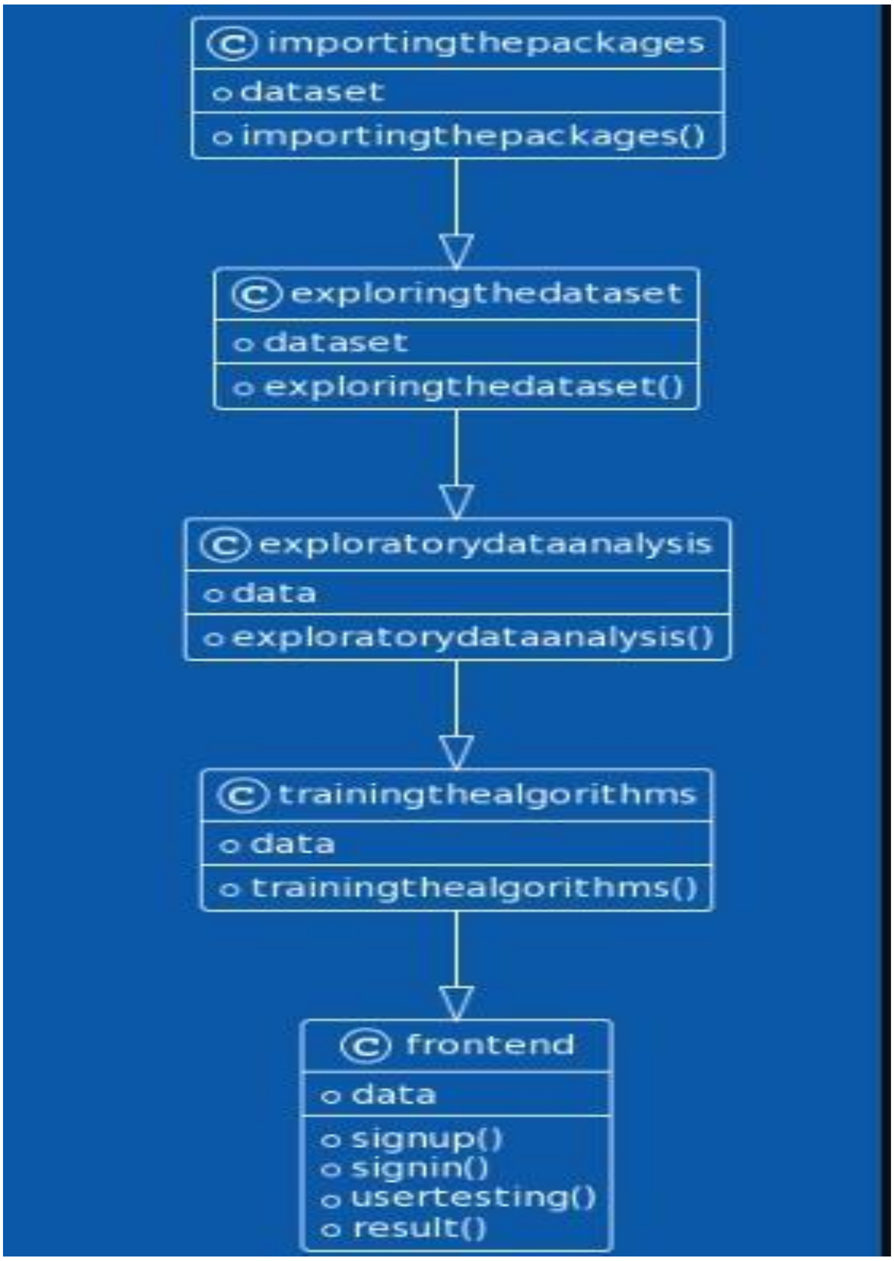

Fig.5.2.2 Class diagram

5.2.3 Activity diagram:

The process flows in the system are captured in the activity diagram. Similar to a state diagram, an activity diagram also consists of activities, actions, transitions, initial and final states, and guard conditions.

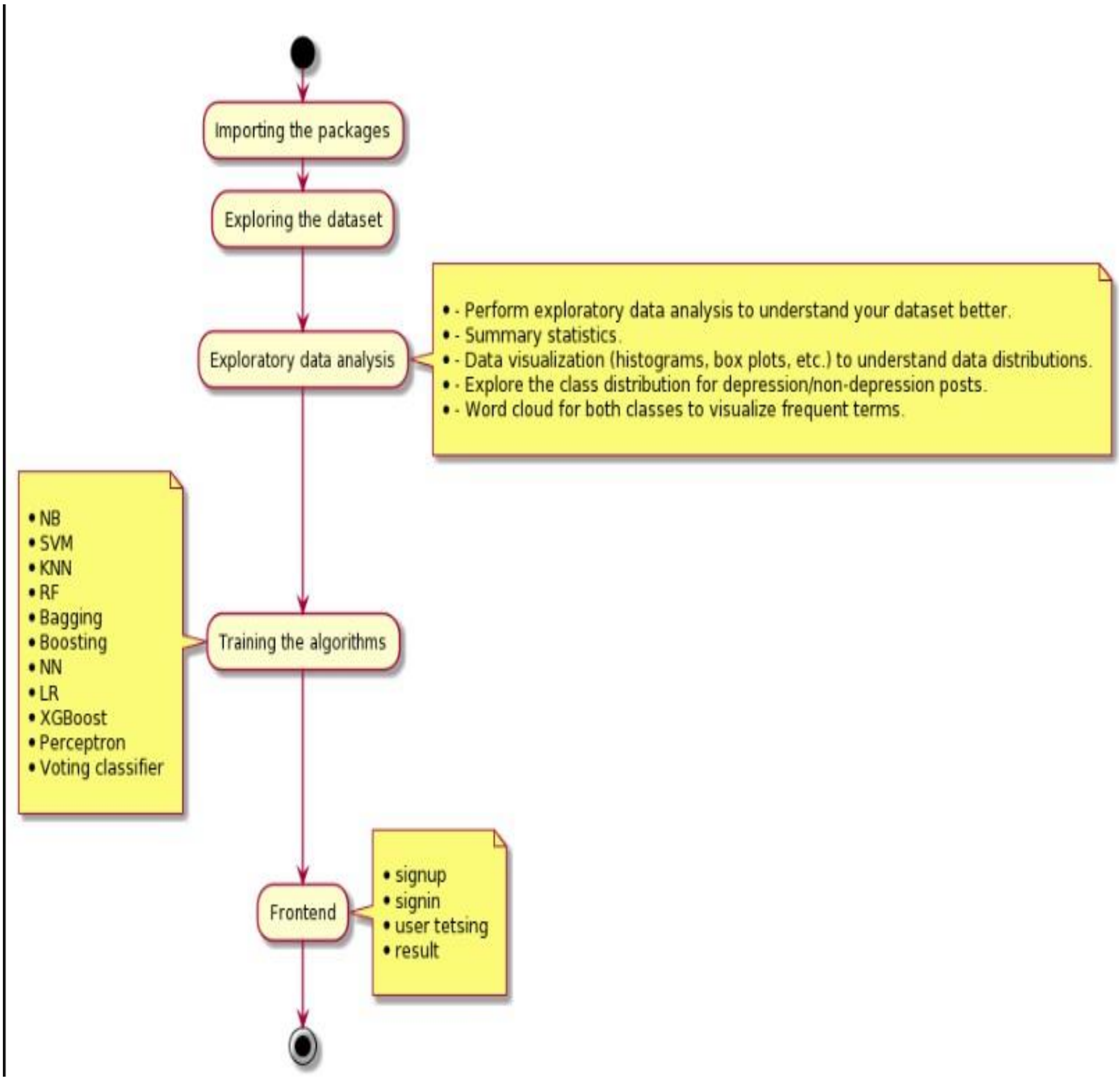

Fig.5.2.3 Activity diagram

5.2.4 Sequence diagram:

A sequence diagram represents the interaction between different objects in the system. The important aspect of a sequence diagram is that it is time-ordered. This means that the exact sequence of the interactions between the objects is represented step by step. Different objects in the sequence diagram interact with each other by passing "messages".

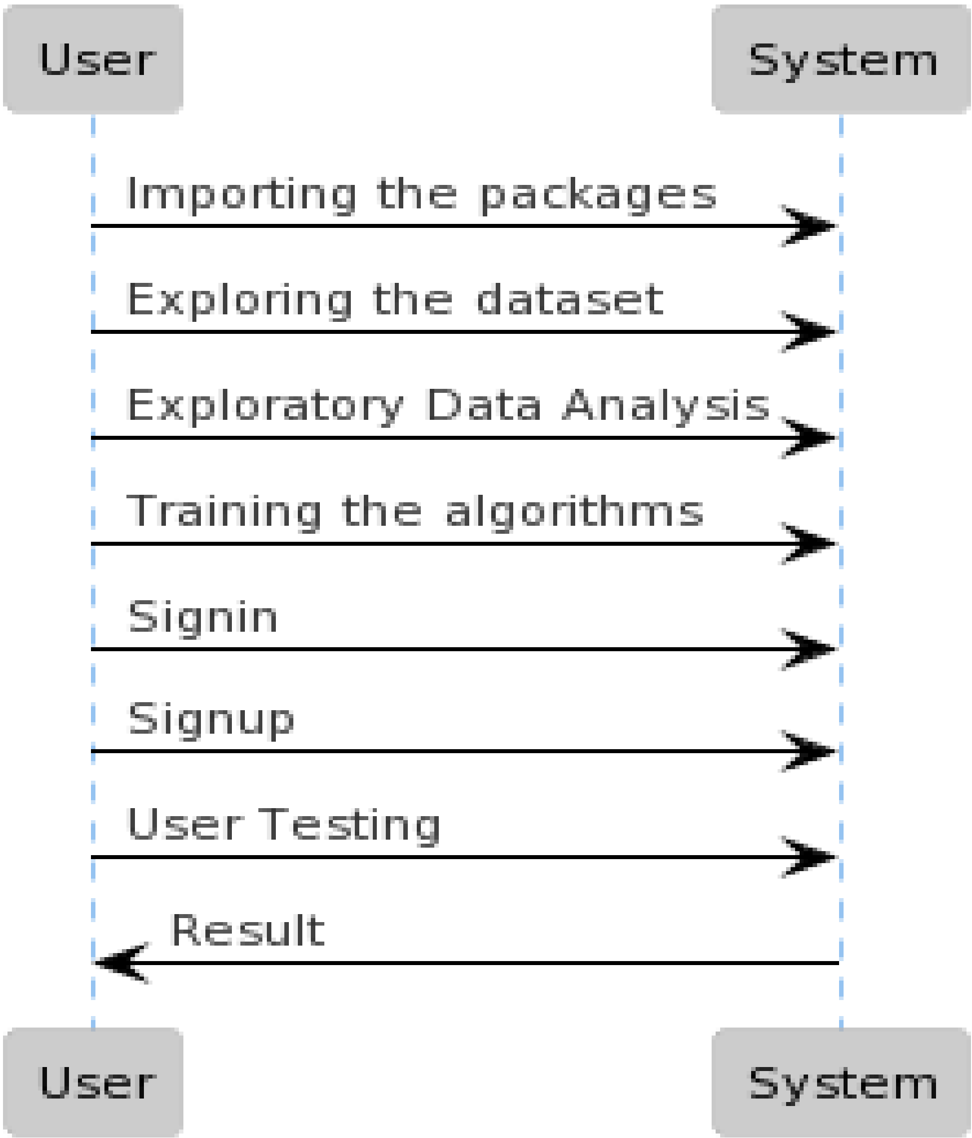

Fig.5.2.4 Sequence diagram

5.2.5 Collaboration diagram:

A collaboration diagram groups together the interactions between different objects. The interactions are listed as numbered interactions that help to trace the sequence of the interactions. The collaboration diagram helps to identify all the possible interactions that each object has with other objects.

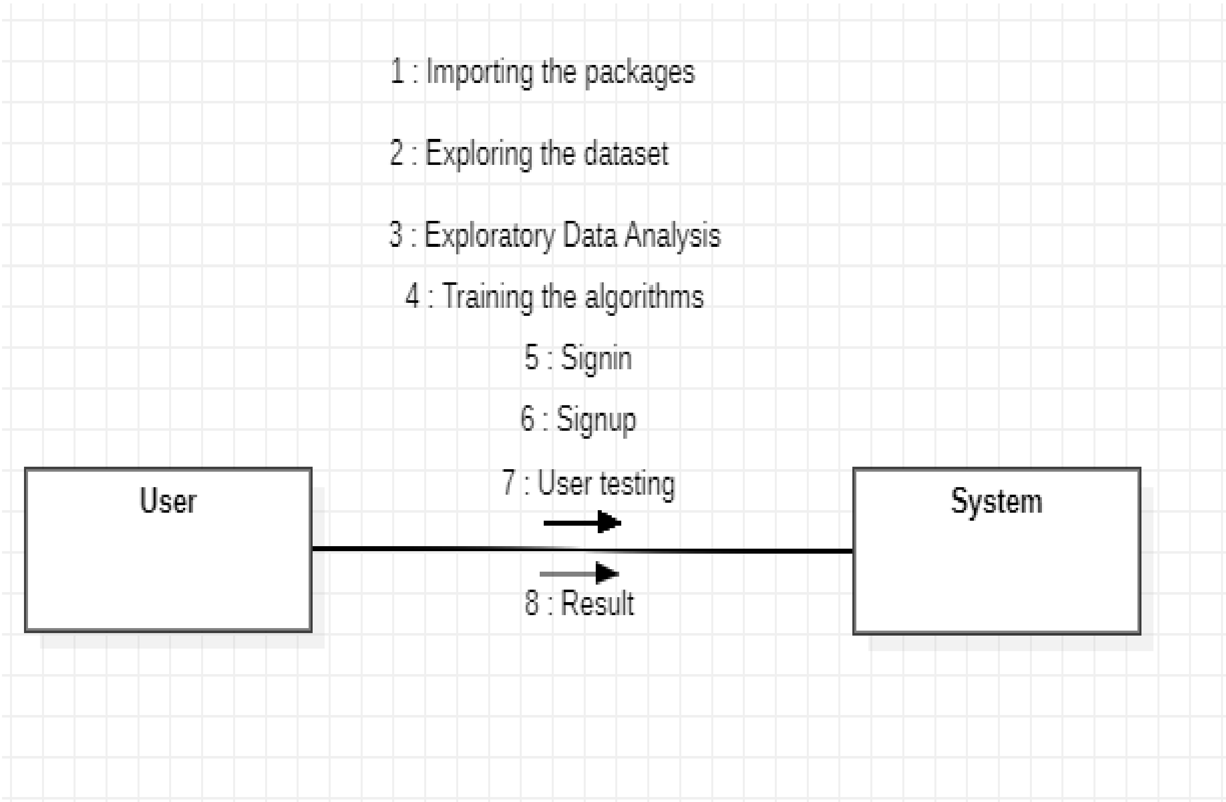

Fig.5.2.5 Collaboration diagram

5.2.6 Component diagram:

The component diagram represents the high-level parts that make up the system. This diagram depicts, at a high level, what components form part of the system and how they are interrelated. A component diagram depicts the components culled after the system has undergone the development or construction phase.

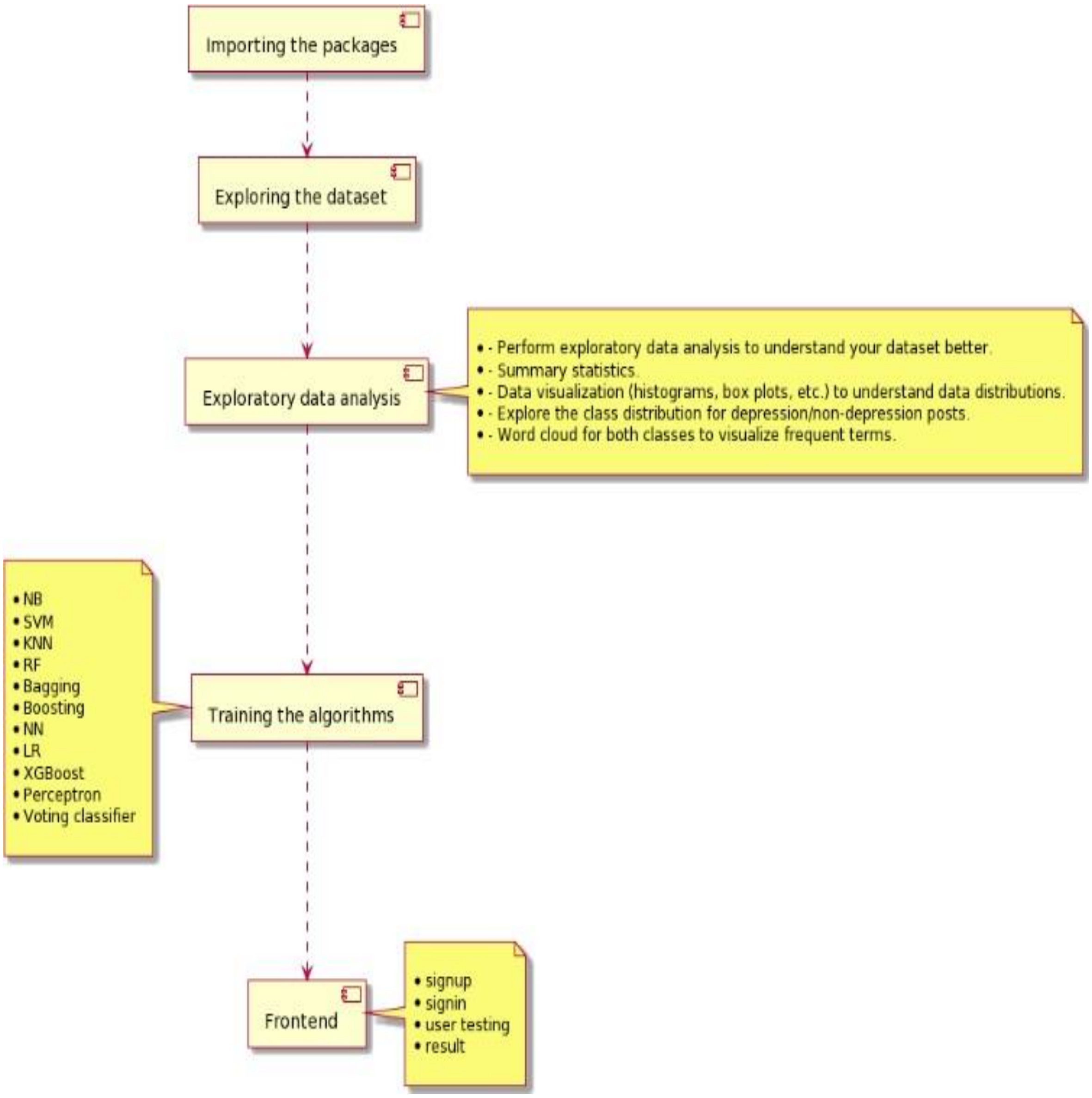

Fig.5.2.6 Component diagram

5.2.7 Deployment diagram:

The deployment diagram captures the configuration of the runtime elements of the application. This diagram is by far most useful when a system is built and ready to be deployed.

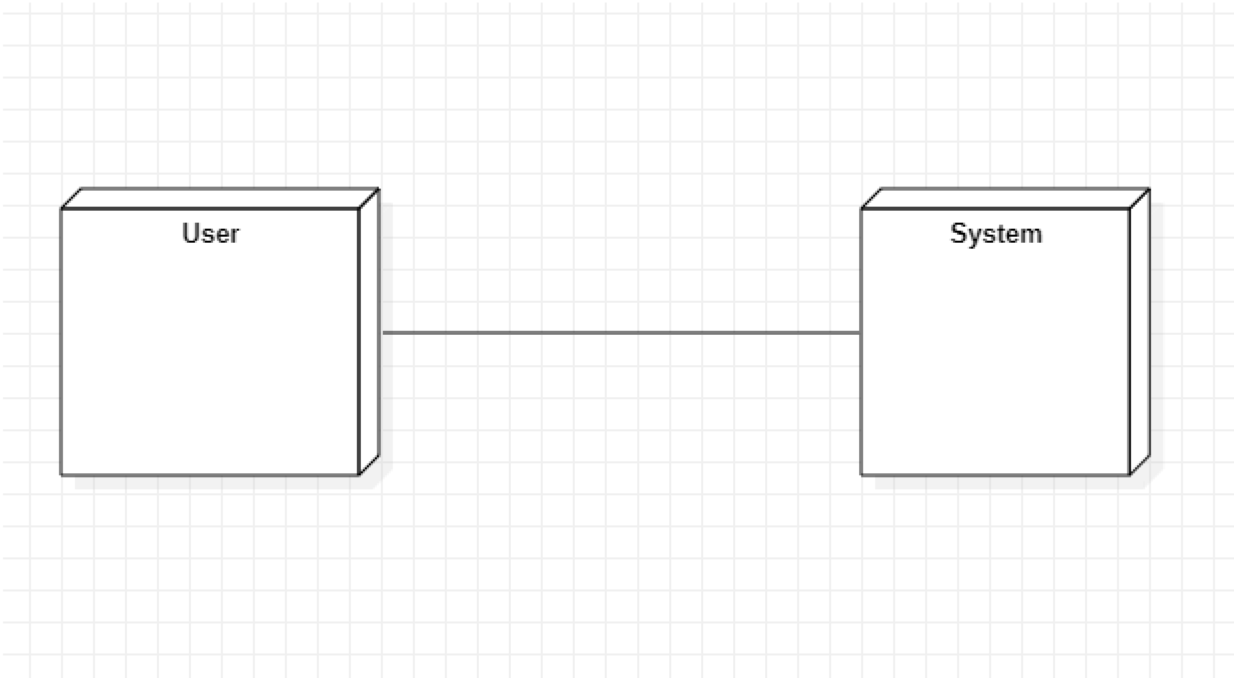

Fig.5.2.7 Deployment diagram

CHAPTER 6

SOFTWARE AND HARDWARE REQUIREMENTS

6.1 SOFTWARE REQUIREMENT

Software requirements deal with defining software resource requirements and prerequisites that need to be installed on a computer to provide optimal functioning of an application. These requirements or prerequisites are generally not included in the software installation package and need to be installed separately before the software is installed.

Platform – In computing, a platform describes some sort of framework, either in hardware or software, which allows software to run. Typical platforms include a computer's architecture, operating system, or programming languages and their runtime libraries.

Operating system is one of the first requirements mentioned when defining system requirements (software). Software may not be compatible with different versions of same line of operating systems, although some measure of backward compatibility is often maintained. For example, most software designed for Microsoft Windows XP does not run on Microsoft Windows 98, although the converse is not always true. Similarly, software designed using newer features of Linux Kernel v2.6 generally does not run or compile properly (or at all) on Linux distributions using Kernel v2.2 or v2.4.

APIs and drivers – Software making extensive use of special hardware devices, like high-end display adapters, needs special API or newer device drivers. A good example is DirectX, which is a collection of APIs for handling tasks related to multimedia, especially game programming, on Microsoft platforms.

Web browser – Most web applications and software depending heavily on Internet technologies make use of the default browser installed on system. Microsoft Internet Explorer is a frequent choice of software running on Microsoft Windows, which makes use of ActiveX controls, despite their vulnerabilities.

1)Anaconda

6.2 HARDWARE REQUIREMENT

The most common set of requirements defined by any operating system or software application is the physical computer resources, also known as hardware, A hardware requirements list is often accompanied by a hardware compatibility list (HCL), especially in case of operating systems. An HCL lists tested, compatible, and sometimes incompatible hardware devices for a particular operating system or application. The following sub-sections discuss the various aspects of hardware requirements.

- **Architecture**

  All computer operating systems are designed for a particular computer architecture. Most software applications are limited to particular operating systems running on particular architectures. Although architecture-independent operating systems and applications exist, most need to be recompiled to run on a new architecture. See also a list of common operating systems and their supporting architectures.

- **Processing power**

  The power of the central processing unit (CPU) is a fundamental system requirement for any software. Most software running on x86 architecture define processing power as the model and the clock speed of the CPU. Many other features of a CPU that influence its speed and power, like bus speed, cache, and MIPS are often ignored.

- **Memory**

  All software, when run, resides in the random access memory (RAM) of a computer. Memory requirements are defined after considering demands of the application, operating system, supporting software and files, and other running processes. Optimal performance of other unrelated software running on a multi tasking computer system is also considered when defining this requirement.

- **Secondary storage**

  Hard-disk requirements vary, depending on the size of software installation, temporary files created and maintained while installing or running the software, and possible use of swap space (if RAM is insufficient).

- **Display adapter**

  Software requiring a better than average computer graphics display, like graphics editors and high-end games, often define high-end display adapters in the system requirements.

- **Peripherals**

  Some software applications need to make extensive and/or special use of some peripherals, demanding the higher performance or functionality of such peripherals.

1) Operating System : Windows Only

2) Processor : i5 and above

3) Ram : 8gb and above

4) Hard Disk : 25 GB in local drive

CHAPTER 7

IMPLEMENTATION

7.1 MODULES:

- Data exploration: using this module we will load data into system
- Processing: Using the module we will read data for processing
- Splitting data into train & test: using this module data will be divided into train & test
- Model generation: Model building - Naive Bayes, SVM, KNN, Random Forest, Bagging, Boosting, Neural Network, Logistic Regression, XGBoost, Perceptron, Voting Classifier, Deep Learning, CNN, LSTM, Transformer Architecture. Algorithms accuracy calculated
- User signup & login: Using this module will get registration and login
- User input: Using this module will give input for prediction
- Prediction: final predicted displayed

7.2 ALGORITHM

Naïve Bayes (NB):

The Naïve Bayes classifier is a supervised machine learning algorithm, which is used for classification tasks, like text classification. It is also part of a family of generative learning algorithms, meaning that it seeks to model the distribution of inputs of a given class or category.

Support Vector Machine (SVM):

Support Vector Machine (SVM) is a powerful machine learning algorithm used for linear or nonlinear classification, regression, and even outlier detection tasks. SVMs can be used for a variety of tasks, such as text classification, image classification, spam detection, handwriting identification, gene expression analysis, face detection, and anomaly detection. SVMs are adaptable and efficient in a variety of applications because they can manage high-dimensional data and nonlinear relationships.

k-nearest neighbors (KNN):

The k-nearest neighbors algorithm, also known as KNN or k-NN, is a non parametric, supervised learning classifier, which uses proximity to make classifications or predictions about the grouping of an individual data point.

Random forest (RF):

Random forest is a commonly-used machine learning algorithm trademarked by Leo Breiman and Adele Cutler, which combines the output of multiple decision trees to reach a single result. Its ease of use and flexibility have fueled its adoption, as it handles both classification and regression problems.

Bagging:

Bagging, also known as bootstrap aggregation, is the ensemble learning method that is commonly used to reduce variance within a noisy dataset. In bagging, a random sample of data in a training set is selected with replacement—meaning that the individual data points can be chosen more than once.

Boosting:

Boosting is a method used in machine learning to reduce errors in predictive data analysis. Data scientists train machine learning software, called machine learning models, on labeled data to make guesses about unlabeled data.

Neural Network (NN):

A neural network is a method in artificial intelligence that teaches computers to process data in a way that is inspired by the human brain. It is a type of machine learning process, called deep learning, that uses interconnected nodes or neurons in a layered structure that resembles the human brain.

Logistic regression (LR):

This type of statistical model (also known as logit model) is often used for classification and predictive analytics. Logistic regression estimates the probability of an event occurring, such as voted or didn't vote, based on a given dataset of independent variables.

XGBoost:

XGBoost is a scalable and highly accurate implementation of gradient boosting that pushes the limits of computing power for boosted tree algorithms, being built largely for energizing machine learning model performance and computational speed.

Voting Classifier:

A Voting Classifier is a machine learning model that trains on an ensemble of numerous models and predicts an output (class) based on their highest probability of chosen class as the output. CNN: Key layers include convolutional layers that use filters, ReLU activation layers to introduce non-linearity, and pooling layers to reduce data size

Long Short-Term Memory (LSTM):

Long Short-Term Memory, is a type of recurrent neural network (RNN) in AI used for processing sequential data like text, speech, and time series.

Transformer Architecture:

Long Short-Term Memory, is a type of recurrent neural network
(RNN) in AI used for processing sequential data like text, speech, and time series.

7.3 SAMPLE CODE:

```
from flask import Flask, render_template, request
import pandas as pd
from sklearn.externals import joblib
import numpy as np
import urllib.request
import urllib.parse

app = Flask(__name__)
model = joblib.load('litemodel.sav')

def sendSMS(apikey, numbers, sender, message):
    data = urllib.parse.urlencode({'apikey': apikey, 'numbers': numbers,'message' : message, 'sender': sender})

        data = data.encode('utf-8')
        request = urllib.request.Request("https://api.textlocal.in/send/?")
        f = urllib.request.urlopen(request, data)
```

```
        fr = f.read()
        return(fr)

defcal(ip):
    input =dict(ip)
    Did_Police_Officer_Attend=input['Did_Police_Officer_Attend'][0]
 age_of_driver = input['age_of_driver'][0]
    vehicle_type= input['vehicle_type'][0]


 age_of_vehicle=input['age_of_vehicle'
 ][0] engine_cc = input['engine_cc'][0]
 day= input['day'][0]
 weather=input['weat
 her'][0] light =
 input['light'][0]
 roadsc =
 input['roadsc'][0]
 gender =
 input['gender'][0]
 speedl =
 input['speedl'][0]
    data=np.array([Did_Police_Officer_Attend,age_of_driver,vehicle_type,age_of_vehi
 cle,  engine_cc, day, weather, roadsc, light, gender, speedl])

 print("logging",d
 ata)data =
 data.astype(float
 )
```

```
data=data.reshap
e(1,-1)
x=np.array([1,3.73,3, 0.69,125,4,1, 1,1,1,30]).reshape(1,-1)

try: result =
model.predict(data)
exceptExceptionase:result=st
r(e)
return str(result[0])


@app.route('/',methods=['GET'])
defindex():
    return render_template('index.html')


@app.route('/visual/',method
s=['GET']) def visual():
    return render_template('visual.html')


@app.route('/sms/',methods
=['POST']) def sms():
    res=cal(reque
    st.form) try:
        resp=sendSMS('UwYs16dD3zM-DKuzZKQYolAJkoba1j0BmRGompsNRs',
'9618205648', 'TXTLCL', 'Severe acceident')
        print (resp)
    exceptExceptionase:
    print(e) return res
```

```
@app.route('/',methods

=['POST']) def get():

    return cal(request.form)



if name_==

    'main':app.run(host='0.0.0.0',debug=

    True,port=4000)

plt.figure(figsize = (20,10))
plt.rcParams.update({'font.size':18})




UT =
np.array([1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29
,30,31,3 2,33,34,35,36])

plt.bar(UT-
        0.2,singleLaneAcc,label='SingleLane',width=0.2,
        align = 'center')
plt.bar(UT,twoLaneAcc,label='TwoLane',width=0.
        2, align = 'center')
plt.bar(UT+0.2,threeLaneAcc,label='ThreeLane',width
        =0.2, align = 'center')
plt.bar(UT+0.4,fourLaneAcc,label='FourLane',width=0
        .2, align = 'center')

plt.xticks(UT,df4['State/UT'],rotation='vertical')
plt.legend(loc = 'best')
plt.title("NumberofACCIDENTSfor1,2,3,4LANEper1Lpopulationofresp.state.")
plt.show()
plt.figure(figsize=(20,10))
plt.rcParams.update({'font.size':18})

plt.bar(UT-0.2,singleInjured,width=0.2,color='b',
        align='center',label='Single Lane')
plt.bar(UT,twoInjured,width=0.2,color='yellow',
        align='center',label='Double lane')
plt.bar(UT+0.2, threeInjured, width=0.2, color='g',
        align='center',label='3Lanesormorew.oMedian')
plt.bar(UT+0.4,fourInjured,width=0.2,color='red',
        align='center',label='4 Lanes with Median')

plt.xticks(UT,df4['State/UT'],rotation='vertical')
```

```
plt.title("NumberofpeopleINJUREDfor1,2,3,4typeoflaneper1Lpopulationofresp.States.")
plt.legend(loc = "best")
plt.show()plt.figure(figsiz
e=(20,10))
plt.rcParams.update({'fon
t.size':18})

plt.bar(UT-0.2,singleLaneKilled,width=0.2,color='b',
        align='center',label='Single Lane')
plt.bar(UT,twoLaneKilled,width=0.2,color='yellow',
        align='center',label='Double lane')
plt.bar(UT+0.2, threeLaneKilled, width=0.2, color='g',
        align='center',label='3Lanesormorew.oMedian')
plt.bar(UT+0.4,fourLaneKilled,width=0.2,color='red',
        align='center',label='4 Lanes with Median')

plt.xticks(UT,df4['State/UT'],rotation='vertical')


plt.title("NumberofpeopleKILLEDfor1,2,3,4LANESper1Lpopulationofresp.States.")
plt.legend(loc = "best")
plt.show()plt.figure(figsiz
e=(20,10))
plt.rcParams.update({'fon
t.size':18})

plt.bar(UT-0.2, singleLaneTotalInjured, width=0.2, align='center',
label='Injured') plt.bar(UT, singleLaneTotalKilled, width=0.2, align='center',
label='Killed')
plt.bar(UT+0.2,singleLaneTotalAccidents,width=0.2,align='center',label="Acci
dents")

plt.xticks(UT,df4['State/UT'],rotation='vertical')
plt.title("NumberofAccidents,peopleKILLED,INJUREDonSINGLELANEper1L
population.")
plt.legend(loc="
best") plt.show()
```

CHAPTER 8

SOFTWARE ENVIRONMENT

8.1 PYTHON LANGUAGE:

Python is an interpreted, object-oriented, high-level programming language with dynamic semantics. Its high-level built in data structures, combined with dynamic typing and dynamic binding, make it very attractive for Rapid Application Development, as well as for use as a scripting or glue language to connect existing components together. Python's simple, easy to learn syntax emphasizes readability and therefore reduces the cost of program maintenance. Python supports modules and packages, which encourages program modularity and code reuse. The Python interpreter and the extensive standard library are available in source or binary form without charge for all major platforms, and can be freely distributed. Often, programmers fall in love with Python because of the increased productivity it provides. Since there is no compilation step, the edit-test-debug cycle is incredibly fast. Debugging Python programs is easy: a bug or bad input will never cause a segmentation fault. Instead, when the interpreter discovers an error, it raises an exception. When the program doesn't catch the exception, the interpreter prints a stack trace. A source level debugger allows inspection of local and global variables, evaluation of arbitrary expressions, setting breakpoints, stepping through the code a line at a time, and so on. The debugger is written in Python itself, testifying to Python's introspective power. On the other hand, often the quickest way to debug a program is to add a few print statements to the source: the fast edit-test-debug cycle makes this simple approach very effective.

Python is a dynamic, high-level, free open source, and interpreted programming language. It supports object-oriented programming as well as procedural-oriented programming. In Python, we don’t need to declare the type of variable because it is a dynamically typed language. For example, x = 10 Here, x can be anything such as String, int, etc.

8.2 Features in Python:

There are many features in Python, some of which are discussed below as follows:

1. Free and Open Source

Python language is freely available at the official website and you can download it from the given download link below click on the Download Python keyword. Download Python Since it is open-source, this means that source code is also available to the public. So you can download it, use it as well as share it.

2. Easy to code

Python is a high-level programming language. Python is very easy to learn the language as compared to other languages like C, C#, Javascript, Java, etc. It is very easy to code in the Python language and anybody can learn Python basics in a few hours or days. It is also a developer-friendly language.

3. Easy to Read

As you will see, learning Python is quite simple. As was already established, Python's syntax is really straightforward. The code block is defined by the indentations rather than by semicolons or brackets.

4. Object-Oriented Language

One of the key features of Python is Object-Oriented programming. Python supports object-oriented language and concepts of classes, object encapsulation, etc.

5. GUI Programming Support

Graphical User interfaces can be made using a module such as PyQt5, PyQt4, wxPython, or Tk in python. PyQt5 is the most popular option for creating graphical apps with Python.

6. High-Level Language

Python is a high-level language. When we write programs in Python, we do not need to remember the system architecture, nor do we need to manage the memory.

7. Extensible feature

Python is an Extensible language. We can write some Python code into C or C++ language and also we can compile that code in C/C++ language.

8. Easy to Debug

Excellent information for mistake tracing. You will be able to quickly identify and correct the majority of your program's issues once you understand how to interpret Python's error traces. Simply by glancing at the code, you can determine what it is designed to perform.

9. Python is a Portable language

Python language is also a portable language. For example, if we have Python code for windows and if we want to run this code on other platforms such as Linux, Unix, and Mac then we do not need to change it, we can run this code on any platform.

10. Python is an Integrated language

Python is also an Integrated language because we can easily integrate Python with other languages like C, C++, etc.

11. Interpreted Language:

Python is an Interpreted Language because Python code is executed line by line at a time. like other languages C, C++, Java, etc. there is no need to compile Python code this makes it easier to debug our code. The source code of Python is converted into an immediate form called bytecode.

12. Large Standard Library

Python has a large standard library that provides a rich set of modules and functions so you do not have to write your own code for every single thing. There are many libraries present in Python such as regular expressions, unit-testing, web browsers, etc.

13. Dynamically Typed Language

Python is a dynamically-typed language. That means the type (for example- int, double, long, etc.) for a variable is decided at run time not in advance because of this feature we don't need to specify the type of variable.

14. Frontend and backend development

With a new project py script, you can run and write Python codes in HTML with the help of some simple tags , , etc. This will help you do frontend development work in Python like javascript. Backend is the strong forte of Python it's extensively used for this work cause of its frameworks like Django and Flask.

15. Allocating Memory Dynamically

In Python, the variable data type does not need to be specified. The memory is automatically allocated to a variable at runtime when it is given a value. Developers do not need to write int y = 18 if the integer value 15 is set to y. You may just type y=18.

8.3 LIBRARIES/PACKGES :-

Tensorflow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

Numpy

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, Numpy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using Numpy which allows Numpy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem.

Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use.

CHAPTER 9

SYSTEM TESTING

System testing, also referred to as system-level tests or system-integration testing, is the process in which a quality assurance (QA) team evaluates how the various components of an application interact together in the full, integrated system or application. System testing verifies that an application performs tasks as designed. This step, a kind of black box testing, focuses on the functionality of an application. System testing, for example, might check that every kind of user input produces the intended output across the application.

Phases of system testing:

A video tutorial about this test level. System testing examines every component of an application to make sure that they work as a complete and unified whole. A QA team typically conducts system testing after it checks individual modules with functional or user-story testing and then each component through integration testing.

If a software build achieves the desired results in system testing, it gets a final check via acceptance testing before it goes to production, where users consume the software. An app-dev team logs all defects, and establishes what kinds and amount of defects are tolerable.

9.1 Software Testing Strategies:

Optimization of the approach to testing in software engineering is the best way to make it effective. A software testing strategy defines what, when, and how to do whatever is necessary to make an end-product of high quality. Usually, the following software testing strategies and their combinations are used to achieve this major objective:

9.1.1 Static Testing:

The early-stage testing strategy is static testing: it is performed without actually running the developing product. Basically, such desk-checking is required to detect bugs and issues that are present in the code itself. Such a check-up is important at the pre-deployment stage as it helps avoid problems caused by errors in the code and software structure deficits.

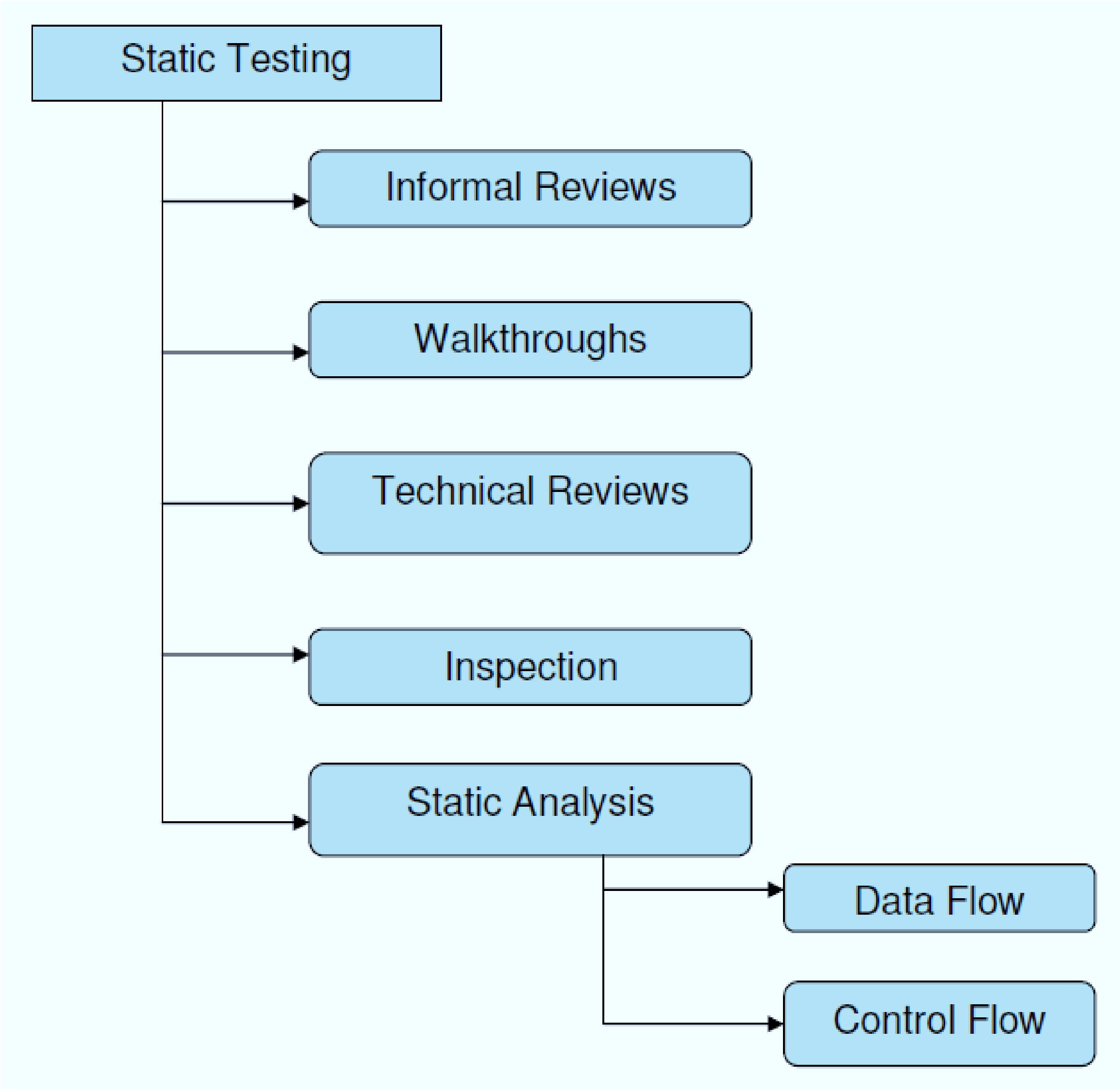

Fig:9.1.1 Static testing

9.1.2 Structural Testing:

It is not possible to effectively test software without running it. Structural testing, also known as white-box testing, is required to detect and fix bugs and errors emerging during the pre-production stage of the software development process. At this stage, unit testing based on the software structure is performed using regression testing. In most cases, it is an automated process working within the test automation framework to speed up the development process at this stage. Developers and QA engineers have full access to the software's structure and data flows (data flows testing), so they could track any changes (mutation testing) in the system's behavior by comparing the tests' outcomes with the results of previous iterations (control flow testing).

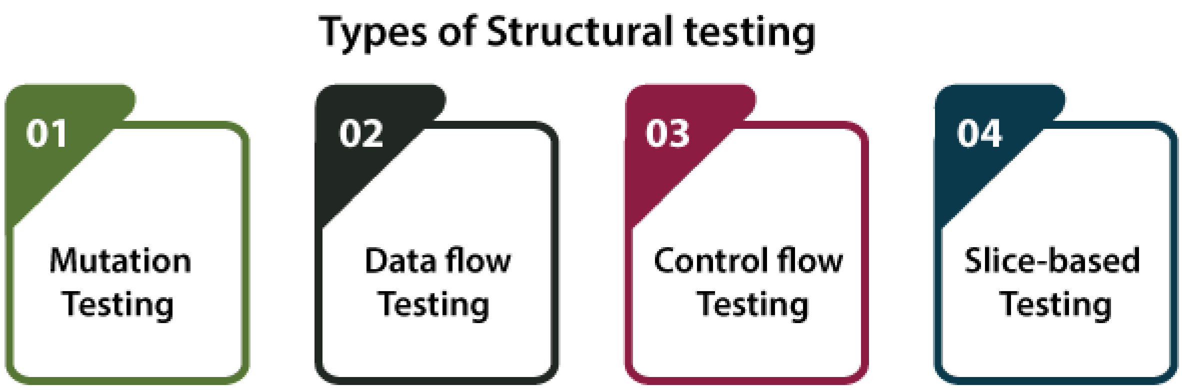

Fig:9.1.2 Types of Structural testing

9.1.3 Behavioral Testing:

The final stage of testing focuses on the software's reactions to various activities rather than on the mechanisms behind these reactions. In other words, behavioral testing, also known as black-box testing, presupposes running numerous tests, mostly manual, to see the product from the user's point of view. QA engineers usually have some specific information about a business or other purposes of the software ('the black box') to run usability tests, for example, and react to bugs as regular users of the product will do. Behavioral testing also may include automation (regression tests) to eliminate human error if repetitive activities are required. For example, you may need to fill 100 registration forms on the website to see how the product copes with such an activity, so the automation of this test is preferable.

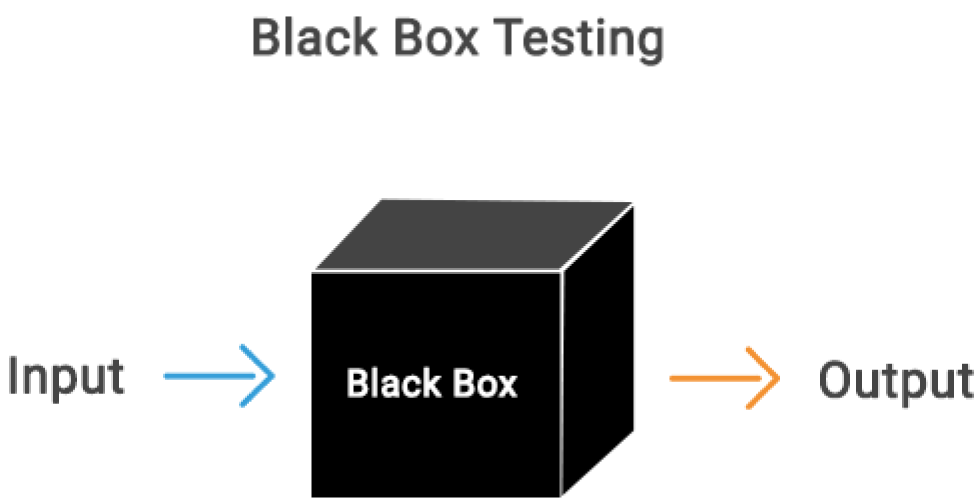

Fig:9.1.3 Behavioral testing

9.2 TEST CASES:

S.NO	INPUT	If available	If not available
1	User signup	User get registered into the application	There is no process
2	User signin	User get login into the application	There is no process
3	Enter input for prediction	Prediction result displayed	There is no process

Table 9.2 Test cases

CHAPTER 10

RESULTS

10.1 LANDING PAGE AND USER SIGN-IN MODULE

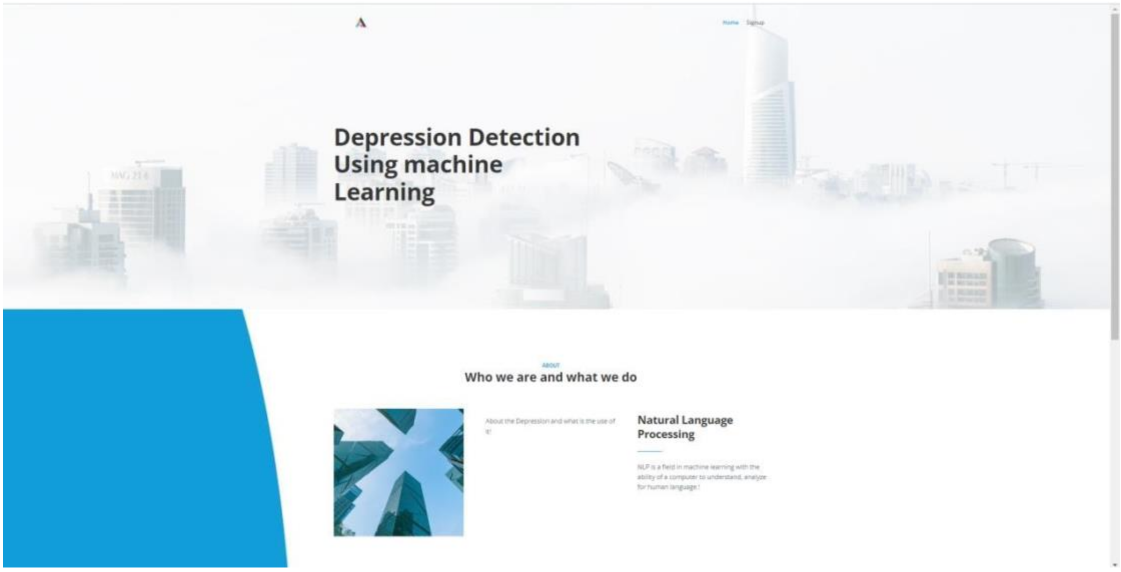

Fig:10.1.1 System landing page

Explanation:

This figure shows the landing page of the proposed Early Depression Detection System. It acts as the initial interface presented to users upon accessing the application. The landing page provides an overview of the system, its purpose, and the available features such as text-based analysis, audio upload, and live voice recording. From this page, users can proceed to authenticate themselves before accessing the depression detection functionalities.

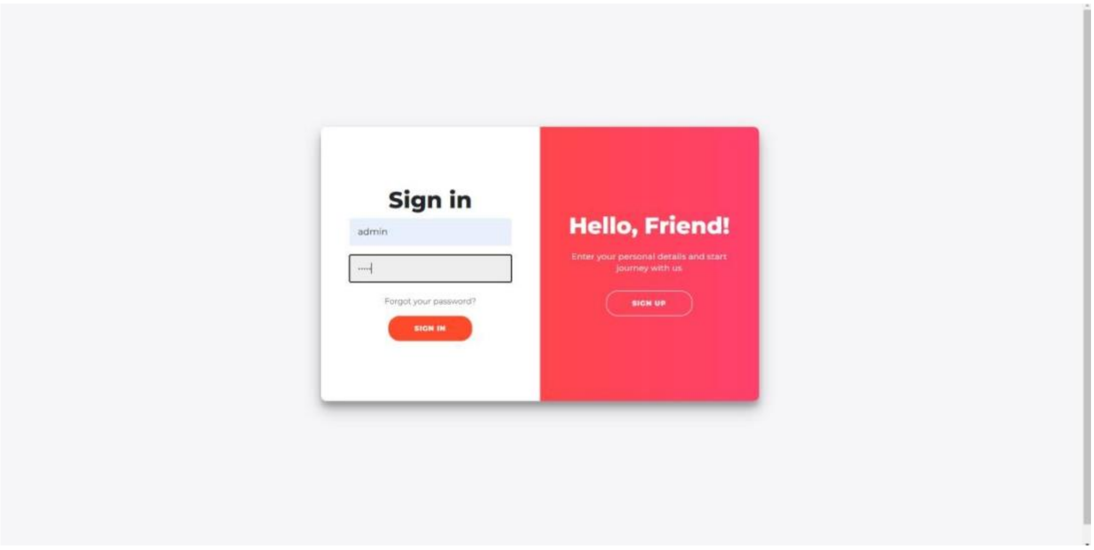

Fig:10.1.2 User Sign-In Page

Explanation:

This figure illustrates the user sign-in interface of the system. Authentication is required to ensure secure access to the depression detection features. Users must provide valid login credentials to proceed. This module helps maintain data privacy and ensures that only authorized users can submit text or audio inputs for analysis.

10.2 Text-Based Depression Detection

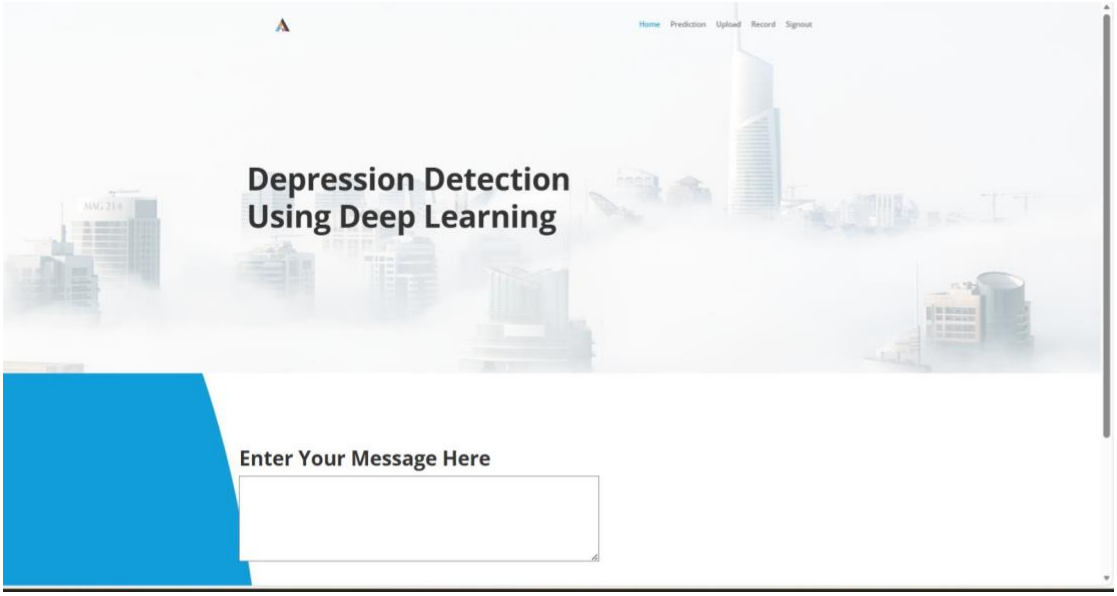

Fig:10.2.1 Text Input Interface for Depression Detection

Explanation:

This figure shows the text input interface of the proposed system, where users enter textual data such as thoughts, feelings, or written expressions. The entered text serves as the primary input for further processing by the deep learning model to identify potential depressive patterns.

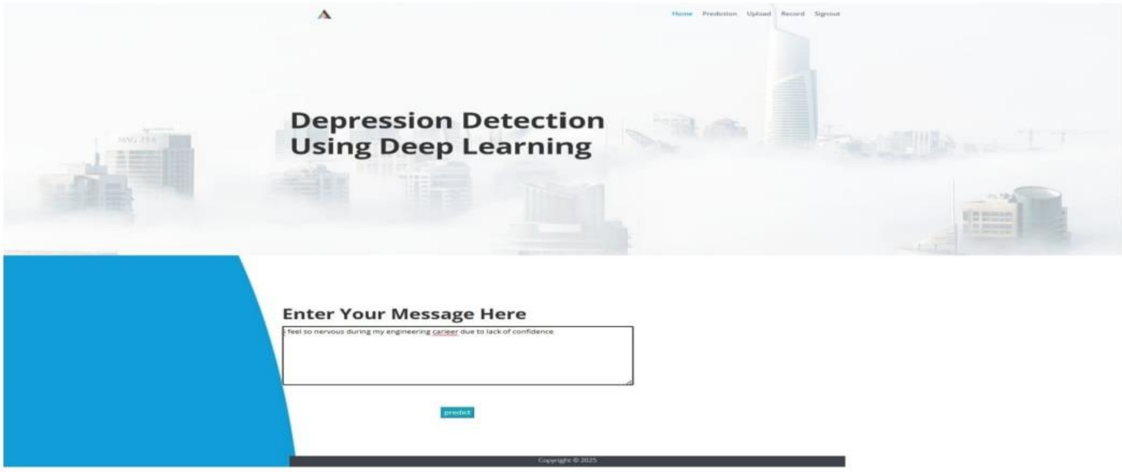

Fig:10.2.2 Text-Based Depression Detection Processing Stage

Explanation:

This figure illustrates the internal processing stage of text-based depression detection. The deep learning model analyzes the input text by extracting emotional, semantic, and linguistic features. Natural language processing techniques are applied to understand sentiment and contextual meaning before classification.

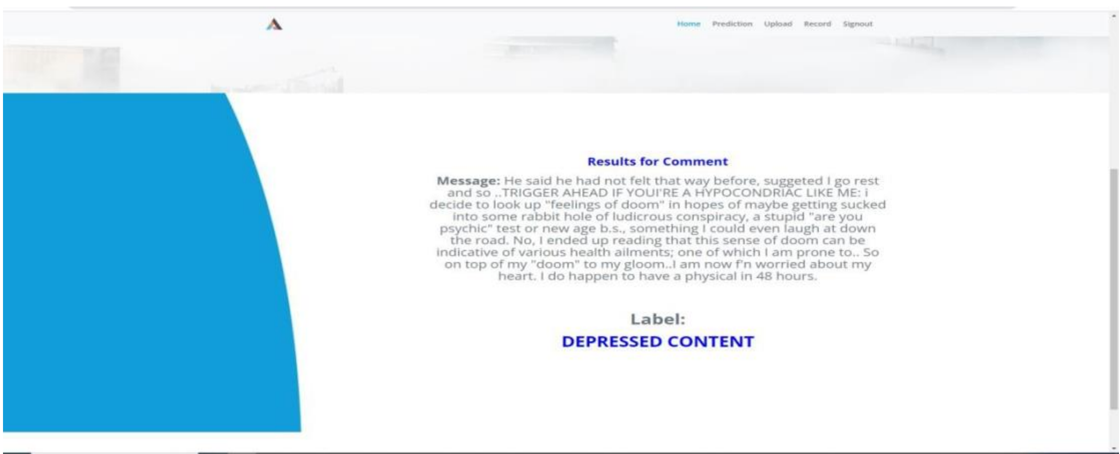

Fig:10.2.3 Text-Based Depression Detection Result

Explanation:

This figure presents the final output of the text-based depression detection module. Based on the extracted features, the deep learning model classifies the input text and displays whether depressive content is detected or not.

10.3 Audio Upload–Based Depression Detection

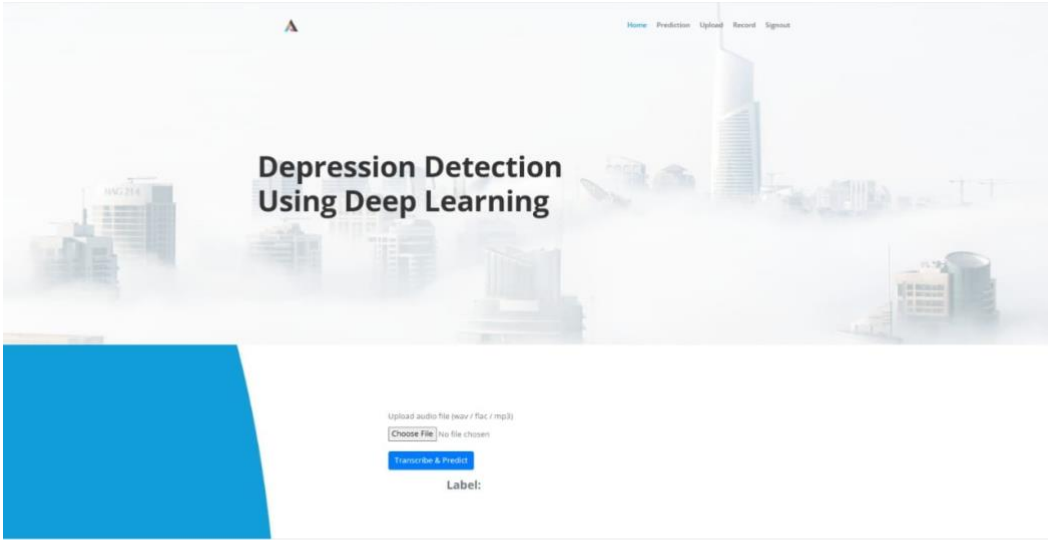

Fig:10.3.1 Audio Dataset Selection for Model Training

Explanation:

This figure illustrates the interface used to upload and select audio datasets. These speech samples are used either for training or testing the proposed depression detection model, enabling the system to learn speech patterns associated with depression.

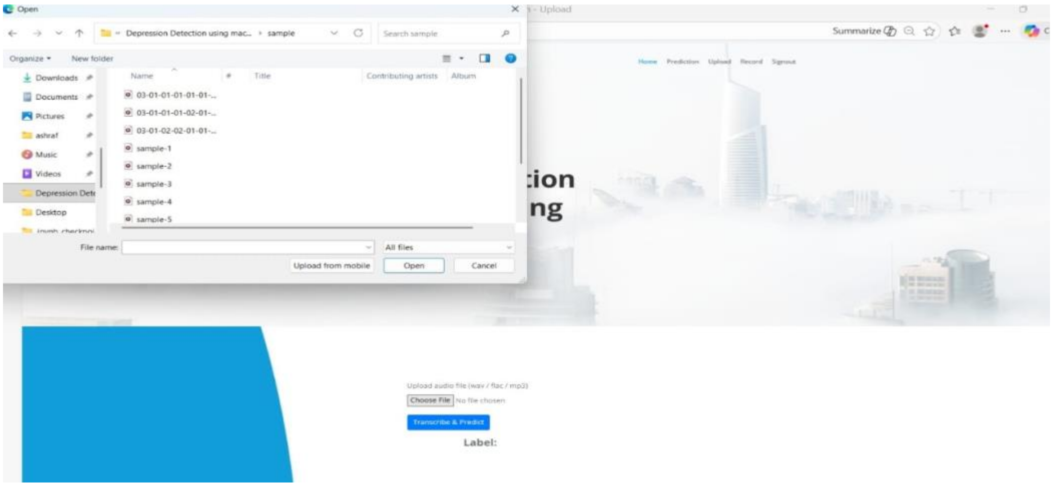

Fig:10.3.2 Transcript Generation from Input Speech Data

Explanation:

This figure shows the conversion of uploaded speech audio into textual transcripts using a speech-to-text mechanism. The generated transcript is further processed for feature extraction and emotional analysis.

Fig:10.3.3 Depression Detection Result Based on Speech Analysis

Explanation:

This figure displays the final prediction result obtained after analyzing the uploaded speech data. The deep learning model evaluates vocal and linguistic features to determine the depression status of the speaker.

10.4 Voice Recording–Based Depression Detection

Fig:10.4.1 Voice Recording Interface for Depression Detection

Explanation:

This figure illustrates the real-time voice recording interface of the system. Users can record their speech directly through the application, which is captured as input for depression analysis.

Fig:10.4.2 Depression Detection Result from Recorded Voice Input

Explanation:

This figure shows the output generated after analyzing the recorded voice input. The system processes the speech signal, extracts relevant acoustic and textual features, and displays the predicted depression status.

CHAPTER 11

CONCLUSION

This project demonstrates the feasibility and effectiveness of applying deep learning and machine learning techniques for early detection of depression. The proposed system follows a systematic methodology consisting of data collection, exploratory data analysis, feature extraction, model building, training, and user interaction through a frontend interface. Deep learning models such as Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN/LSTM), and transformer-based architectures were implemented and evaluated, along with conventional machine learning classifiers including Naive Bayes, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest, Bagging, Boosting, Logistic Regression, XGBoost, Perceptron, and Voting Classifier. The experimental results show that deep learning models perform better in capturing complex patterns present in the data when compared to traditional methods. The system is capable of classifying user inputs into depression and non-depression categories, thereby supporting early screening and preliminary mental health assessment. Furthermore, the use of an existing sentiment analysis dataset helps address limitations related to data availability, cost, and time. Overall, the work carried out in this project provides a reliable and efficient approach that can assist mental health professionals in early diagnosis and timely intervention.

CHAPTER 12

FUTURE SCOPE

This project is to further enhance the system by incorporating additional modalities such as facial expressions, voice signals, and behavioral patterns in order to improve the accuracy of depression detection. The system can be extended to support real-time analysis using live voice inputs, textual conversations, and continuous user interactions. Training the models on larger, multilingual, and more diverse datasets can improve robustness and generalization across different user populations. The adoption of advanced deep learning architectures, particularly improved transformer-based models and attention mechanisms, may further enhance contextual understanding and prediction performance. In addition, deploying the system as a secure web-based or mobile application can increase accessibility and practical usability. Future work may also focus on clinical validation, strengthening ethical considerations, and ensuring data privacy and security to support real-world deployment of the system.

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